

Intuitive Inference: Enhancing Post-Double Selection Inference on Treatment Effects with Intuition-Based Penalties

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Abstract

This thesis examines the ‘Post-Double Selection’ methodology introduced by Belloni et al. (2014b) for achieving valid inference on treatment effects in high-dimensional datasets. While Belloni et al. (2014b) and Wüthrich and Zhu (2023) suggest the ex-post addition of economically intuitive variables to reduce bias, they do not provide a robust methodology for this integration. This study addresses this gap by incorporating economically intuitive variables directly into the selection process, overcoming the limitations of traditional ex-post additions for confounding variable selection, particularly in high-noise environments. Sufficient conditions are established for asymptotically valid inference, allowing imperfect beliefs when the size of the amelioration set is controlled. Monte Carlo simulations demonstrate that this integrated approach significantly enhances treatment effect estimation, showing marked improvement over the traditional Post-Double Selection method under certain conditions. An empirical re-examination of the effect of abortion on crime rates, as studied by Donohue and Levitt (2001), highlights the practical benefits of this approach, leading to a more comprehensive control selection and meaningful economic interpretations.

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The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

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1 Introduction

Albert Einstein is widely quoted as saying, “The intuitive mind is a sacred gift, and the rational mind is a faithful servant.” However, in contemporary econometric research, this perspective is undergoing a reversal. Since the introduction of the first simple econometric variable selection algorithm, the stepwise regression, the practice of variable selection has shifted from rational economic reasoning and intuition towards complex statistical models (Desboulets, 2018). Nowadays, most explanatory variable selection in simple and complex models tends to be guided more by minimizing a loss function or a metric than by economic intuition. In forecasting, the shift toward econometric rationality is driven by the pursuit of improved in-sample and out-of-sample results, wherein model selection aims solely to enhance accuracy (Berrevoets et al., 2024; Chernozhukov et al., 2018). Major economic institutes and government agencies increasingly rely on black-box mechanisms to uncover and exploit the benefits of increasingly high-dimensional data. For instance, researchers at the ECB are using random forest models to forecast inflation (Chinn et al., 2023; Lenza et al., 2023), and researchers at the IMF are using neural networks to forecast tail risk (Sakurai & Chen, 2024).

The shift to machine learning methods is emphasized by the current combinatorial explosion in empirical econometric practice, where the (baseline) control set is enriched with larger sets of interactions and other complex transformations. This is done to leverage higher computational power and model flexibility to achieve potential efficiency gains (Wüthrich & Zhu, 2023). Although modern machine learning models are designed to identify only those covariates with the most potent predictive ability, these approaches, often driven by regularization, induce model sparsity well, provided a sparse linear form can effectively capture the outcome. Empirically, this is often the case, resulting in more accurate forecasts with smaller variances (Belloni et al., 2014a). However, as regularization introduces bias, these machine learning models need to be refined to exploit the potential information in high-dimensional data and provide better-informed answers to causal questions, such as the inference of causal or treatment effects, while minimizing their side effects (Belloni & Chernozhukov, 2011). Hence, there is a continuous strong drive to further develop high-dimensional causal econometric models in the academic and practitioner fields.

A recent approach in this discussion is the ‘Post-Double Selection’, which employs a double selection step for asymptotically valid inference on treatment effects in a high-dimensional environment (Belloni et al., 2014a). Empirical results demonstrate significant reductions in bias through the use of a regularization model that includes a double (sparsity-inducing) selection step. The methodology relies on conducting two preliminary Lasso regressions: one of the outcome variable on the control set and another of the treatment on the control set. The main objective is to choose sufficient confounding controls (or simply controls) such that conditioning on these selected controls renders the treatment variable exogenous (Belloni et al., 2017). A final Ordinary Least Squares (OLS) regression of the outcome variable on the treatment and the selected controls from the preliminary step substantially reduces bias in estimating the treatment effect compared to a single selection step and similar variants. Following this methodology, the Post-Double Selection provides valid uniform inference across a broad range of models.

At first sight, the Post-Double Selection method appears to be yet another data-driven

mechanism where minimizing a loss function yields an 'optimal' set of controls, and the economic intuition behind control selection is discarded. However, in the literature, there is precedent for reintroducing economic intuition into the model. Researchers are even recommended to “*always augment the union of the selected controls with an ‘amelioration set’ of controls motivated by economic theory and prior knowledge*” (Wüthrich & Zhu, 2023). For the purpose of this thesis, following Desboulets (2018), economically intuitive variables are defined as raw variables with a clear economic interpretation and are typically expected to act as confounding variables for the relevant causal problem. The addition of economically intuitive controls ex-post for the final OLS regression is reasoned by arguing that combining a formal, rigorous approach with complementary economic intuition can significantly enhance the estimation of treatment effects, especially when the treatment is exogenous to the controls (Belloni et al., 2014a). Furthermore, even when this assumption does not hold, the amelioration set is seen as a potentially significant addition for ensuring valid inference by conditioning on the necessary controls, thereby making the treatment plausibly exogenous (Belloni et al., 2017).

In the existing literature, the inclusion of the amelioration set is typically limited to an ex-post addition to the final OLS regression (Belloni et al., 2014a; Gillen et al., 2014; Wüthrich & Zhu, 2023). However, this simplistic approach disregards proper selection criteria and lacks robustness. Under unfavourable conditions, the ex-post addition of the amelioration set could even worsen causal inference. If the controls in the amelioration set are not described by the Data Generating Process (DGP), this could lead to increased variance. Additionally, multicollinearity may arise if the controls in the amelioration set are strongly correlated with previously selected controls, thereby increasing the standard deviations of parameter estimates. Whilst the treatment effect is not directly affected, the precision of any post-hoc economic interpretation based on the coefficient estimates is affected.

Empirically, the necessity for an amelioration set is especially high in problems with noisy data, where the true effects of fundamental variables are overshadowed by noise. In such a setting, perfect recovery of the real representation of the DGP, even asymptotically, cannot longer be assumed (Wainwright, 2019). As a result, the Post-Double Selection method could yield a subset in which complex transformed variables dominate over economically intuitive variables, potentially better fitting the noisy data. If these controls are chosen through a spurious relationship, their addition could prevent the detection of important confounding variables within the Post-Double Selection and cause bias by missing underlying trends (Belloni et al., 2016). In this context, the amelioration set can mitigate the bias due to the missing confounding variables, albeit against a higher variance. This thesis, therefore, aims to provide an integrated alternative to the ex-post addition of the amelioration set by applying lower penalization to economically intuitive variables within Post-Double Selection Lasso regressions. This leads to the following research question:

To what extent can integrated amelioration set penalization be used in Post-Double Selection to reduce bias and improve inference in a high-dimensional dataset with many controls?

With respect to existing literature, the main findings of this thesis are interesting for several reasons. First, different implementations of data-dependent and data-independent amelioration

set penalties are proposed, producing competitive inferential results for treatment effects in approximately sparse models. These methods are more robust than those currently described in the literature as the information on the amelioration set is integrated throughout the selection procedure instead of added ex-post, exploiting available information to fine-tune the penalization weights. Additionally, the methods accommodate both homoscedastic and heteroscedastic error distributions and are tested across various levels of residual system noise. Second, several data-dependent and data-independent amelioration set penalties are evaluated through Monte Carlo simulations, yielding a better treatment effect estimation than Post-Double Selection under certain conditions. This differential is particularly significant when the treatment variable and its confounding variables are highly correlated and the relationship is subject to low residual noise levels. This finding is applied in an empirical replication examining the effect of abortion on crime rates, which features such a data structure (Donohue & Levitt, 2001). The most basic amelioration set penalty, which removes penalization of the amelioration set in the Post-Double Selection, acts as a hybrid between regular Post-Double Selection and Post-Double Selection with the ex-post added amelioration set: an intermediate number of economically intuitive controls and an intermediate total number of controls are selected, yielding a methodology that provides a solid middle ground between economic interpretation and econometric robustness.

The thesis is structured as follows: Section 2 provides a brief overview of the current literature on Post-Double Selection. Section 3 covers the relevant methodological background and establishes sufficient conditions for valid inference with the amelioration set penalties. The proof is given in Appendix A. Section 4 discusses the data used in the research. Next, Section 5 describes and explains the steps required to perform valid usage of the Post-Double Selection and the amelioration set penalties. Section 6 presents the results of the Monte Carlo simulations and offers a practitioner’s recommendation for using the amelioration set penalties. Section 7 applies these findings to an empirical case. Finally, Section 8 presents the conclusions drawn, the limitations of the research, and ideas for future research.

Notation. $\text{supp}(b)$ denotes the set of indices of the non-zero elements of b , with cardinality $|\text{supp}(b)| = \|b\|_0$. The average expectation operator is defined as $\mathbb{E}_n[f] := \mathbb{E}_n[f(\cdot)] := \sum_{i=1}^n f(\cdot)/n$. The norms used are the ℓ_2 -norm, denoted by $\|\cdot\|_2$, the ℓ_1 -norm, denoted by $\|\cdot\|_1$, and the ℓ_∞ -norm, denoted by $\|\cdot\|_\infty$, representing the largest absolute element of a vector.

2 Related Literature

On a fundamental level, this thesis builds directly on the existing literature of machine learning for causal inference, first introduced by Belloni, Chernozhukov and Hansen (2013). In this seminal paper, the Post-Double Selection method is introduced with the Lasso estimator as a means of achieving approximate sparsity. An extension paper provides a formal proof for uniformly valid inference on treatment effects after selection in a setting with high-dimensional controls (Belloni et al., 2014a). Further papers extend the double-Lasso method to various applications, such as high-dimensional panel models (Belloni et al., 2016), high-dimensional approximately sparse quantile regression models (Belloni et al., 2019), and principled variable selection (Urminsky et al., 2019). Advancements generalize the impact of regularization bias and

overfitting of the parameter of interest using Neyman-orthogonal moments and efficient data-splitting, in a method known as *debiased-ML*, of which the Post-Double Selection is a specific case (Chernozhukov et al., 2018, 2022). All of these papers mention ℓ_1 -regularization, specifically the Lasso estimator, as a good means to approximate sparsity. This thesis will primarily focus on the original methodology of the Post-Double Selection and extend the literature by providing a robust improvement to the control selection.

2.1 Data-Driven Penalization

The ℓ_1 -regularization parameter plays a critical role in the Lasso estimator, balancing the tradeoff between variance and bias. Too many unnecessary controls may be included if the penalty is not conservative enough, leading to higher variance. Conversely, a penalty that is too conservative can increase omitted variable bias. Bickel et al. (2009) first proposed a choice for the Lasso penalty based on Gaussian error assumptions and data dimensionality. Building on this, Belloni and Chernozhukov (2013) introduced data-independent penalty loadings for Post-Lasso under Gaussian errors, and Belloni et al. (2012) introduced data-driven penalty loadings accommodating non-Gaussian errors. Finally, Belloni, Chernozhukov and Hansen (2013) developed an algorithm to estimate the residual variance in the data, providing a framework for optimal penalty loadings under non-Gaussian, heteroscedastic errors. The thesis will extend the methodology for non-Gaussian homo- and heteroscedastic errors by accommodating beliefs based on economic intuition, adding an extra dimension to the estimation of penalty loadings.

In addition, an important feature of data-driven penalties is their co-dependence in determining optimal loadings: each loading is fit to its control based on all available information. Therefore, an adjusted loading for one control will affect the loading of another. Therefore, this thesis provides a useful insight into the behaviour of data-driven penalty loadings under beliefs, which are not necessarily data-driven but do translate to lower penalization.

2.2 Practitioner’s Recommendations

Considering the existing literature on Post-Double Selection, the theoretical focus of Post-Double Selection is often overshadowed by diverse empirical applications.¹ Belloni et al. (2014a) stands out as one of the few theoretical papers that proposes practical recommendations for using causal inference estimators across data conditions. They analyze the performance of Post-Lasso and Post-Double Selection estimators against the infeasible Oracle benchmark across various combinations of first and second-stage R^2 values² and coefficient designs, to guide the appropriate use of Post-Double Selection in causal inference analysis. The authors find that Post-Lasso performs poorly across all combinations of R^2 and designs, managing only to control bias when the treatment is uncorrelated with the control (first-stage $R^2 = 0$). Post-Lasso also generally performs better in a scenario where the confounding variables in the first-stage differ from those in the second-stage regression, occasionally even outperforming Post-Double Selection

¹See for instance, Qiu et al. (2020) for an epidemiological application on the transmission of Covid-19, Dhar et al. (2022) for an economic application on adolescent gender attitudes, and Hangartner et al. (2021) for a psychological application on hiring discrimination.

²Referring to the correlation between the treatment variable and the design matrix, and between the outcome and the design matrix, respectively.

under conditions of low first- and second-stage R^2 . This situation arises when fundamental variables are significantly distorted by noise and the identities of the confounding variables are uncertain. Amelioration penalties aim to precisely address this challenge by providing supplementary information to better capture confounding variables under low first- and second-stage R^2 . Otherwise, the Post-Double Selection estimator performs similarly to the Oracle across all combinations of R^2 and achieves its best performance under high first-stage R^2 when the identities of the confounding variables are well known. This thesis aims to extend recommendations for practitioners by conducting an updated analysis using amelioration penalties across different designs and error distributions. In addition, this thesis will consider varying DGPs where the confounding variables in the first stage are not the same as those in the second stage.

2.3 Bayesian Lasso

The idea of imposing a lower level of penalization for controls that are considered fundamental can be explored through the lens of the Bayesian Lasso, introduced by Park and Casella (2008). The paper suggests that the behaviour of the Lasso estimator can be interpreted through a Bayesian posterior distribution, assuming that the priors on the regression parameters follow independent LaPlace distributions. More specifically, the Lasso penalty estimates are equivalent to the mode of the posterior distributions. The shape of the LaPlace distribution, peaking sharply at the origin, produces a similar shrinkage effect as regularization with ℓ_1 -penalty, and in practice, the solutions of both methods should be identical. Therefore, the results of this thesis can be extended to Bayesian statistics by translating adjusted penalties in the amelioration set into different shaped priors for their respective Bayesian counterparts. For instance, the prior for a variable that is believed to be economically intuitive, under discrete amelioration set penalties, can be modelled as a discrete posterior distribution with $\mathbb{P}(X = 0) = 1$. Although this thesis is not directly related to the Bayesian Lasso, exploring this connection could be an interesting topic for future research.

3 Theoretical Framework

This section provides an overview of the relevant theoretical background necessary to understand the choices made for the penalties in the amelioration set. First, the theory behind approximately sparse models and its contextualization to causal inference is presented. Next, the Lasso, Post-Lasso, indirect Post-Lasso, and Post Double Selection estimators are introduced alongside the relevant penalties to ensure validity. Lastly, sufficient conditions for the inferential validity of the amelioration set penalties are stated, followed by a formal result whose proof is given in Appendix A.

3.1 Approximately Sparse Models

In an econometric model where the number of regressors p is relatively large compared to the number of observations n , conducting a simple OLS regression is impossible. However, if a small subset of regressors s can approximately capture the main features of the regression, identifying this subset and performing regression using only these regressors offers a feasible approach to

OLS regression (Belloni & Chernozhukov, 2011). This intuition behind a model that can be approximated by a small subset of its regressors describes the structure of an ‘*Approximately Sparse Model*’ (ASM) and motivates the consideration of regularization techniques to identify s . Approximate sparsity requires that a small number s of regressors approximate the regression function accurately; the number of candidate controls p may exceed n .

The following model is introduced to provide intuition into the approximation of a high-dimensional model through an ASM (Belloni, Chernozhukov & Hansen, 2013):

$$y_i = f(z_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n, \quad (3.1)$$

where y_i is the outcome variable, z_i is a vector of primary regressors that are significant drivers of the regression function, and ε_i are i.i.d. errors. The regression function $f(z_i)$ is unknown, implying that the exact relationship between z_i and y_i is also unknown. To approximate the regression function, the design matrix $X = [x_1, \dots, x_n]'$ is introduced, where each $x_i = P(z_i)$. Here, $P(z_i)$ represents a chosen collection of transformations of z_i , possibly involving constants, splines, interactions, or powers. The coefficients determining their contribution to the regression function remain unknown. The resulting approximation of the approximately sparse model through the design matrix is as follows:

$$y_i = x_i' \beta_0 + r_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (3.2)$$

where β_0 represents the coefficients and r_i the approximation errors. Assuming there exists a set of unknown regressors s such that a linear combination of these regressors produces sufficiently low approximation errors, the model is considered approximately sparse. This equivalence is characterized by a linear combination of regressors with $|\text{support}(\beta_0)| \leq s \ll p$, resulting in relatively small r_i . However, identifying the specific elements of β_0 is challenging, and researchers typically introduce a high-dimensional design matrix where $p \gg n$ in an effort to uncover these complex underlying relationships.³ In practice, pursuing the sparse identification of a high-dimensional model is only beneficial if it is believed that y_i can be well-approximated using a much smaller subset of regressors (Belloni & Chernozhukov, 2011).

3.2 Approximately Sparsity for Inference

This approach of identifying the driving regressors in models with high-dimensional controls using approximately sparse models can be extended to causal inference (Belloni et al., 2014a). Consider the following partially linear model:

$$y_i = d_i' \alpha_0 + g(z_i) + u_i, \quad \mathbb{E}[u_i | z_i, d_i] = 0, \quad (3.3)$$

$$d_i = m(z_i) + v_i, \quad \mathbb{E}[v_i | z_i] = 0, \quad (3.4)$$

where y_i is the outcome variable, d_i is the treatment variable, α_0 is the coefficient of interest capturing the treatment effect, z_i are the controls, and u_i and v_i are zero-mean errors. The

³Consider the empirical case in Section 7, where a baseline control set of seven regressors expands to include 284 non-multicollinear controls. In the end, however, only an average of 10 controls were selected in the Post-Double Selection.

functions $g(\cdot)$ and $m(\cdot)$ are unknown and describe the relationships of the controls on the outcome variable and the treatment variable, respectively. Similar to $f(z_i)$ in (3.2), the unknown nature and potentially complex structure of these functions hinder the ability to identify the relevant controls in the partially linear model. The conditional-mean restrictions in the model provide the identifying assumptions. Sparse approximations are used to control the influence of z_i when estimating the treatment effect under these assumptions. In practice, uncovering the identities of z_i is challenging, and sparse approximations provide a method to approximate $g(z_i)$ and $m(z_i)$. Consider the following sparse approximation of the partially linear model (Belloni et al., 2014a):

$$y_i = d_i' \alpha_0 + \underbrace{x_i' \beta_{g0}}_{g(z_i)} + r_{gi} + u_i, \quad (3.5)$$

$$d_i = \underbrace{x_i' \beta_{m0}}_{m(z_i)} + r_{mi} + v_i, \quad (3.6)$$

where $x_i' \beta_{g0}$ and $x_i' \beta_{m0}$ are the sparse approximations for $g(z_i)$ and $m(z_i)$, and r_{gi} and r_{mi} are the respective approximation errors. Conditional on sufficiently small approximation errors and other regularity conditions, some of which are discussed in Section 3.6, valid causal inference can be performed on the treatment effect α_0 (Belloni et al., 2014a). Therefore, the challenge remains to find a feasible and valid method to construct such an approximately sparse model given a high-dimensional design matrix.

3.3 One-stage Lasso Estimation

The Lasso estimator provides a simple, theoretically guaranteed, and computationally feasible solution to induce sparsity through ℓ_1 -regularization. The estimator shrinks some coefficients and sets others to zero without requiring prior knowledge of the structure of the DGP (Tibshirani, 1996):

$$(\hat{\alpha}_L, \hat{\beta}_L) \in \arg \min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \mathbb{E}_n \left[(y_i - d_i \alpha - x_i' \beta)^2 \right] + \frac{\lambda}{n} \|\beta\|_1, \quad (3.7)$$

where $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$, and λ is a specified penalty level. The treatment coefficient is unpenalised whenever the treatment enters a regression. Although ℓ_1 -regularization helps Lasso reduce overfitting, the shrinkage of coefficients to zero can introduce significant bias. The Post-Lasso estimator attempts to alleviate Lasso's shrinkage bias by performing an OLS regression on the variables selected from an initial Lasso selection. Removing the penalty from the final regression can reduce shrinkage bias, while post-Lasso prediction rates remain controlled under suitable conditions (Belloni & Chernozhukov, 2013):

$$(\hat{\alpha}_{PL}, \hat{\beta}_{PL}) \in \arg \min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \mathbb{E}_n \left[\left(y_i - d_i \alpha - \sum_{j=1}^p x_{i,j} \beta_j \right)^2 \right] : \beta_j = 0 \text{ if } \hat{\beta}_{L,j} = 0, \forall j. \quad (3.8)$$

The Indirect Post-Lasso is an adaptation of the Post-Lasso, conditioning on the indirect effect of d_i on $x_{i,j}$ in the first-stage estimation to reduce the possibility of missing confounding variables. Next, an OLS is performed on the union of the controls selected in the first stage and the

treatment variable (Belloni & Chernozhukov, 2011):

$$\hat{\beta}_d = \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}_n \left[(d_i - x_i' \beta)^2 \right] + \frac{\lambda}{n} \|\beta\|_1, \quad (3.9)$$

$$(\hat{\alpha}_{IPL}, \hat{\beta}_{IPL}) \in \arg \min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \mathbb{E}_n \left[\left(y_i - d_i \alpha - \sum_{j=1}^p x_{i,j} \beta_j \right)^2 \right] : \beta_j = 0 \text{ if } \hat{\beta}_{d,j} = 0, \forall j. \quad (3.10)$$

This estimator should intuitively recognize the confounding variables better, shrinking the OVB more effectively than the Post-Lasso. However, if the correlation between d_i and $x_{i,j}$ is negligible, and the correlation between y_i and $x_{i,j}$ is high, the treatment effect might become difficult to identify. Therefore, the single-stage Lasso might benefit from a second-stage estimation to address this pitfall.

3.4 Two-stage Lasso Estimation

Although the Lasso estimator is restricted to estimating a single model at a time, it can be used sequentially to accommodate partial linear models. For instance, the Post-Double Selection estimator, which sparsely approximates the partial linear model described in (3.3) and (3.4), is based on a two-stage estimation: a first-stage Lasso estimation of d_i on $x_{i,j}$ followed by a second-stage Lasso estimation of y_i on $x_{i,j}$. A final OLS regression is then performed of y_i on d_i and the union of the selected variables from the first- and second-stage Lasso estimations (Belloni et al., 2014a):

$$\hat{\beta}_x = \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}_n \left[(y_i - x_i' \beta)^2 \right] + \frac{\lambda}{n} \|\beta\|_1. \quad (3.11)$$

The second selection approximates the conditional mean of the outcome given the controls. From the partially linear model,

$$\ell(z_i) := \mathbb{E}[y_i | z_i] = \alpha_0 m(z_i) + g(z_i), \quad e_{li} := y_i - \ell(z_i) = \alpha_0 v_i + u_i. \quad (3.12)$$

Therefore, the noise relevant for this selection includes the residual variation in the treatment as well as the structural disturbance. This distinction will matter when specifying the baseline penalties.

Let $\hat{I}_1$ denote the controls selected by (3.9) and $\hat{I}_2$ denote the controls selected by (3.11). Let the union be $\hat{I}$. The final OLS regression is as follows:

$$(\hat{\alpha}, \hat{\beta}) \in \arg \min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \mathbb{E}_n [(y_i - \alpha d_i - x_i' \beta)^2], \quad b_j = 0 \quad (j \notin \hat{I}), \quad (3.13)$$

This procedure can be understood through Frisch–Waugh–Lovell partialling out. The final coefficient is obtained by regressing the outcome residual on the treatment residual after both have been projected on the selected controls (Heij et al., 2004). Under the identifying restrictions of the model, the two selections help control the approximation bias in this coefficient. They do not imply exact recovery of α_0 in a finite sample. The conditions for asymptotically valid

inference are given in Section 3.6.

The main strength of the two-stage Post Double Selection estimation over a single-stage Lasso or Post-Lasso lies in the additional estimations of d_i on $x_{i,j}$ and y_i on $x_{i,j}$. Intuitively, there are two scenarios in which the two-stage estimation may improve on the one-stage estimation. First, if d_i and $x_{i,j}$ are strongly correlated but y_i and $x_{i,j}$ are not, the single-stage estimation might fail to capture this relationship and not condition on the confounding $x_{i,j}$ due to the low correlation with the outcome variable, whereas the two-stage estimation would account for this through the first-stage estimation. Second, if both y_i and $x_{i,j}$, as well as d_i and $x_{i,j}$, are strongly correlated, the single-stage estimation might fail to distinguish the treatment effect from $x_{i,j}$, whereas the two-stage estimation would account for this through the second-stage estimation by estimating y_i on $x_{i,j}$ without conditioning on d_i .

3.5 Lasso-Penalization

So far, in the theoretical framework, the regularization behaviour of the Lasso estimator has only depended on the value of λ . However, differences in the scale of the controls and the variability of the noise scores can be accounted for through an additional parameter, Ψ . In this section, the theory behind the penalty choices is provided, along with an introduction to Ω , the parameter that regulates the amelioration set penalization. Together, these parameters allow the baseline regularization to account for the noise in the regression, while incorporating beliefs about the controls through a lower level of penalization.

3.5.1 λ -Penalty: Regularization

In the context of forecasting, cross-validation is a popular method for choosing λ , as it aims to minimise prediction error. For inference on the treatment effect, the main function of λ in the theoretical analysis is to dominate the ‘effective noise’ in the system while maintaining the regularization bias sufficiently small (Belloni & Chernozhukov, 2011; Belloni et al., 2014a). Furthermore, λ regulates the sparsity in the Lasso estimation through the additional cost of non-zero coefficients. Therefore, increasing the penalty to control the noise must be balanced against the regularization bias introduced by shrinking the coefficients, such that the selected controls can still provide a sufficiently accurate approximation of the regression function.

The following definitions will improve clarity in the derivation of the penalty. Let $q_{mi} = d_i$ and $q_{\ell i} = y_i$ denote the dependent variables in the two selections, with conditional means $f_m = m$ and $f_\ell = \ell$, respectively. Their disturbances are $e_{mi} = v_i$ and $e_{\ell i} = \alpha_0 v_i + u_i$, as given in (3.12). For either regression $r \in \{m, \ell\}$, write $q_r = (q_{r1}, \dots, q_{rn})'$ and $e_r = (e_{r1}, \dots, e_{rn})'$. The criterion function of the Lasso estimator can then be written in sample-average and matrix form as

$$\widehat{Q}_r(b) = \mathbb{E}_n[(q_{ri} - x_i' b)^2] = \frac{(q_r - Xb)'(q_r - Xb)}{n}. \quad (3.14)$$

Differentiating the criterion and evaluating at β_{r0} gives

$$\begin{aligned}\nabla \widehat{Q}_r(b) &= -\frac{2}{n}X'(q_r - Xb), \\ S_r &:= \nabla \widehat{Q}_r(\beta_{r0}) = -\frac{2}{n}X'(e_r + r_r) = -2\mathbb{E}_n[x_i e_{ri}] - 2\mathbb{E}_n[x_i r_{ri}],\end{aligned}\tag{3.15}$$

where $r_r = (r_{r1}, \dots, r_{rn})'$ collects the approximation errors $r_{ri} = f_r(z_i) - x_i' \beta_{r0}$. The first term, $S_r^e = -2\mathbb{E}_n[x_i e_{ri}]$, is the noise score. The second term arises because the sparse representation need only approximate the conditional mean. It disappears in an exactly sparse model; otherwise, it is controlled separately in the proof through the approximation-error bound. Following Belloni and Chernozhukov (2013), the penalty is chosen to dominate the maximum absolute noise score with probability tending to one:

$$\frac{\lambda_r}{n} \geq c \|S_r^e\|_\infty = 2c \|\mathbb{E}_n[x_i e_{ri}]\|_\infty, \quad c > 1.\tag{3.16}$$

The constant c allows the penalty to exceed the noise score by a fixed proportion. Taking the maximum across controls makes the inequality apply to every coordinate of the noise score simultaneously. This is the quantity that the penalty must control, rather than the noise in any single observation.

To obtain a feasible penalty, first consider conditionally Gaussian, homoscedastic disturbances, $e_r \mid X \sim \mathcal{N}(0, \sigma_r^2 I_n)$. Let $g = (g_1, \dots, g_n)'$ be a vector of independent standard normal draws, generated independently of X . The conditional distribution of the noise score then satisfies

$$n \|S_r^e\|_\infty = 2 \|X' e_r\|_\infty \stackrel{d|X}{=} \sigma_r \Lambda, \quad \Lambda := 2 \|X' g\|_\infty = 2n \|\mathbb{E}_n[x_i g_i]\|_\infty.\tag{3.17}$$

This is an equality in conditional distribution under the stated Gaussian assumption. Let $\Lambda(1 - \gamma \mid X)$ denote the conditional $(1 - \gamma)$ -quantile of Λ , where $0 < \gamma < 1$ is the tail probability. With the disturbance standard deviation known, choosing $\lambda_r = c\sigma_r \Lambda(1 - \gamma \mid X)$ gives

$$\mathbb{P}\left(\frac{\lambda_r}{n} \geq 2c \|\mathbb{E}_n[x_i e_{ri}]\|_\infty \mid X\right) \geq 1 - \gamma.\tag{3.18}$$

The quantile can be approximated by repeatedly drawing g and computing Λ for the observed design matrix. Replacing σ_r by an estimate gives the feasible penalty

$$\widehat{\lambda}_r = c\widehat{\sigma}_r \Lambda(1 - \gamma \mid X).\tag{3.19}$$

The score bound then also requires control of the estimation error in $\widehat{\sigma}_r$; (3.18) does not automatically hold after this substitution. This distinction is discussed by Belloni and Chernozhukov (2013, Section 2.4). The scale estimation used in this thesis is described in Section 5.

The standard Lasso estimation with this penalty can therefore be written as

$$\widehat{\beta}_{\Lambda, r} \in \arg \min_{b \in \mathbb{R}^p} \left\{ \frac{(q_r - Xb)'(q_r - Xb)}{n} + \frac{c\widehat{\sigma}_r}{n} \Lambda(1 - \gamma \mid X) \|b\|_1 \right\}.\tag{3.20}$$

The latter notation is useful as $\Lambda(1-\gamma \mid X)$, the X -dependent penalty quantile, remains constant given a fixed regressor matrix and tail probability. The full penalty can still change when $\hat{\sigma}_r$ is updated. Furthermore, the quantile accounts for the dependence between the controls. For instance, if the columns are normalised such that $\mathbb{E}_n[x_{ij}^2] = 1$, a Gaussian union bound gives

$$\Lambda(1-\gamma \mid X) \leq 2\sqrt{n} \Phi^{-1}\left(1 - \frac{\gamma}{2p}\right), \quad (3.21)$$

where Φ is the standard normal distribution function. The bound on the right does not use the correlations in X , whereas the simulated quantile does. The X -dependent choice can therefore be less conservative (Chernozhukov et al., 2016).

For asymptotic score control, γ can be chosen to tend to zero, while c remains fixed above one. The Gaussian construction illustrates the penalty choice in this special case. The theorem below instead states the required score and loading conditions explicitly, allowing other disturbance distributions when these conditions hold. A construction satisfying them under an additional tail restriction is given in Appendix A.3; the general conditions are stated in Appendix A.1.

3.5.2 Ψ -Penalty: Equivariance

The penalty can be adjusted to the variability of the individual noise scores through $\hat{\Psi}_r = \text{diag}(\hat{\psi}_{r1}, \dots, \hat{\psi}_{rp})$, where the elements are positive penalty loadings. This adjustment is particularly relevant under heteroscedasticity, as the variability of $x_{ij}e_{ri}$ can differ across controls. The resulting Lasso estimator is as follows (Belloni et al., 2014a):

$$\hat{\beta}_{\Psi,r} \in \arg \min_{b \in \mathbb{R}^p} \left\{ \hat{Q}_r(b) + \frac{\lambda_r}{n} \left\| \hat{\Psi}_r b \right\|_1 \right\}, \quad \left\| \hat{\Psi}_r b \right\|_1 = \sum_{j=1}^p \hat{\psi}_{rj} |b_j|. \quad (3.22)$$

Each coefficient is therefore penalized according to its respective loading, such that the noise scores in (3.16) are standardised coordinate by coordinate. With the inclusion of $\hat{\Psi}_r$, the required score condition can be written in three equivalent forms:

$$\begin{aligned} \frac{\lambda_r}{n} \hat{\psi}_{rj} &\geq 2c |\mathbb{E}_n[x_{ij}e_{ri}]| \quad \text{for every } j \leq p, \\ \frac{\lambda_r}{n} &\geq 2c \max_{j \leq p} \frac{|\mathbb{E}_n[x_{ij}e_{ri}]|}{\hat{\psi}_{rj}}, \\ \frac{\lambda_r}{\sqrt{n}} &\geq 2c \max_{j \leq p} \frac{|n^{-1/2} \sum_{i=1}^n x_{ij}e_{ri}|}{\hat{\psi}_{rj}}. \end{aligned} \quad (3.23)$$

To see why these loadings standardise the scores, suppose the observations are independent and $\mathbb{E}[e_{ri} \mid x_i] = 0$. Let $\Sigma_{e,r}(X)$ be the diagonal matrix with elements $\mathbb{E}[e_{ri}^2 \mid X]$. The conditional score covariance is

$$\begin{aligned} V_r(X) &:= \text{Var}\left(\frac{X'e_r}{\sqrt{n}} \mid X\right) = \frac{X'\Sigma_{e,r}(X)X}{n}, \\ [V_r(X)]_{jj} &= \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_{ij}e_{ri} \mid X\right) = \frac{1}{n} \sum_{i=1}^n x_{ij}^2 \mathbb{E}[e_{ri}^2 \mid X]. \end{aligned} \quad (3.24)$$

Dividing a score by $[V_r(X)]_{jj}^{1/2}$ therefore gives it conditional variance one, provided this variance is positive. Since the disturbances are unobserved, preliminary residuals $\hat{e}_{ri}$ are used to construct the feasible loadings

$$\hat{\psi}_{rj} = \left\{ \frac{1}{n} \sum_{i=1}^n x_{ij}^2 \hat{e}_{ri}^2 \right\}^{1/2}, \quad j = 1, \dots, p, \quad (3.25)$$

as in Belloni et al. (2014a) and Chernozhukov et al. (2016). The square root is needed because the loadings estimate standard deviations. Equivalently, their squares are the diagonal elements of $X' \text{diag}(\hat{e}_{r1}^2, \dots, \hat{e}_{rn}^2) X / n$.

These loadings also explain the equivariance property. If a control x_{ij} is multiplied by a non-zero constant a , its loading is multiplied by $|a|$. Replacing its coefficient by b_j/a then leaves both the fitted value and the penalty contribution unchanged. Thus, with the same preliminary residuals and penalty level, changing the units of a control does not change the fitted regression.

The residual-based loadings already include the disturbance scale. A common scale factor can be moved from the loadings to λ_r without changing their product in the objective, as in the normalisation used in the methodology. Standardising the scores does not itself make them Gaussian or establish the score bound in (3.23). Validity also depends on the preliminary residuals and the choice of λ_r . The theorem therefore maintains the score condition explicitly, together with positive lower and upper bounds on the baseline loadings.

3.5.3 Ω -Penalty: Amelioration set

The amelioration set penalties are introduced through a $(p \times 1)$ vector Ω_r that scales the penalization of individual controls. Let $A = \hat{I}_3$ denote the indices of the prespecified amelioration set, with $|A| \leq k$. For each of the two selection regressions, the weights satisfy

$$0 \leq \omega_{rj} \leq 1 \quad (j \in A), \quad \omega_{rj} = 1 \quad (j \notin A), \quad (3.26)$$

where $\Omega_r = (\omega_{r1}, \dots, \omega_{rp})'$. Let $D_{\Omega,r} = \text{diag}(\omega_{r1}, \dots, \omega_{rp})$ be the corresponding diagonal matrix. The Lasso estimation with the amelioration set penalties can then be written as

$$\begin{aligned} \hat{\beta}_r &\in \arg \min_{b \in \mathbb{R}^p} \left\{ \frac{(q_r - Xb)'(q_r - Xb)}{n} + \frac{\lambda_r}{n} \left\| \hat{\Psi}_r D_{\Omega,r} b \right\|_1 \right\}, \\ \hat{\beta}_r &\in \arg \min_{b \in \mathbb{R}^p} \left\{ \mathbb{E}_n[(q_{ri} - x_i' b)^2] + \frac{\lambda_r}{n} \sum_{j=1}^p \hat{\psi}_{rj} \omega_{rj} |b_j| \right\}. \end{aligned} \quad (3.27)$$

The controls outside A receive the same baseline penalty as in (3.22), while the controls within A receive a lower penalty according to their respective weights. Although the penalties outside the amelioration set remain unchanged, their selection can still be affected by the lower penalization of the controls within the set. In particular, allowing a control to enter the regression more easily changes the residual variation available to the other controls, such that the resulting selection also depends on the relationships within the design matrix.

The weights may differ between the two selections and may use information from the same sample. The identities of the controls in A are specified before estimation, while the values

assigned to their weights can be data-dependent. For each minimisation in (3.27), the loadings and weights are held fixed. Let $\hat{I}_m = \text{supp}(\hat{\beta}_m)$ and $\hat{I}_\ell = \text{supp}(\hat{\beta}_\ell)$ denote the resulting selections, corresponding to $\hat{I}_1$ and $\hat{I}_2$ in (3.13). The final regression is performed on d_i and their union, $\hat{I} = \hat{I}_m \cup \hat{I}_\ell$.

For the Discrete penalty, $\omega_{rj} = 0$ is imposed on the amelioration set, while the Half-Discrete penalty sets these weights to one-half. In the case of the Discrete penalty, the controls in A are accounted for before the remaining controls are selected, as shown through partialling out in Appendix A.4. Therefore, integrating the amelioration set within the Lasso estimation can affect the selection of the remaining controls, whereas adding the same set ex-post takes place after these controls have already been selected.

Lower penalization may help to retain economically intuitive controls that would otherwise be excluded, although incorrect beliefs can also lead to the inclusion of irrelevant controls. Therefore, the conditions in the following section account for the size of the amelioration set, including controls that do not contribute to the sparse approximation. The specific values of Ω_r under the different beliefs will be further explored in the methodology.

3.6 Conditions for Valid Inference

Valid inferential results for treatment effects in ASMs depend on the approximate sparsity and rate conditions established in Belloni et al. (2014a). The approximate sparsity condition requires that the functions $m(z_i)$ and $g(z_i)$ in (3.3) and (3.4) can be sufficiently well approximated using a small subset of the full control set, while the rate condition regulates the growth of this subset relative to the number of observations. For the amelioration set penalties, the inclusion of beliefs within the Lasso estimations raises the question of whether these conditions continue to hold when some controls receive lower penalization. In particular, an imperfect amelioration set may introduce controls that do not contribute to the sparse approximation, such that its size must also be accounted for in the rate condition.

The following conditions are given for the partially linear model in (3.3) and (3.4), maintaining the conditional-mean restrictions used to identify the treatment effect. Observations are assumed to be i.i.d., with sufficiently bounded moments of the disturbances and non-vanishing residual variation in the treatment, allowing both heteroscedastic and non-Gaussian errors. Let $s \geq 1$ denote a bound on the number of controls in the union of the two sparse approximations. The approximate sparsity condition is as follows:

$$|\text{supp}(\beta_{m0}) \cup \text{supp}(\beta_{g0})| \leq s, \quad \|r_m\|_{L_4} + \|r_g\|_{L_4} \leq C\sqrt{s/n}, \quad (3.28)$$

where $\|r_f\|_{L_4} = \{\mathbb{E}[r_{fi}^4]\}^{1/4}$ and C is an absolute constant. Compared to a mean-square approximation condition, (3.28) also restricts the fourth moments of the approximation errors. This additional restriction allows the errors to be controlled in the derivation of both the treatment-effect estimator and its standard error. Under exact sparsity, $r_{mi} = r_{gi} = 0$, such that the approximation condition is immediately satisfied.

The penalty and design conditions ensure that the Lasso selections produce sufficiently accurate approximations using a sufficiently small number of controls. Following the discussion in

Section 3.5, the baseline penalty must dominate the empirical noise scores, while the design matrix must allow small subsets of controls to be distinguished from one another. These conditions are imposed before the amelioration weights are applied. The controls outside the amelioration set therefore retain the regularization used in the usual Lasso argument, while the k controls within the set may receive lower penalties, including a penalty of zero. Accounting for these additional controls gives the following rate condition:

$$\frac{(s+k)\log(p \vee n \vee 3)}{\sqrt{n}} \rightarrow 0, \quad (3.29)$$

where $\vee$ denotes the maximum. The condition restricts the combined growth of the sparse approximation and the amelioration set, such that controls receiving lower penalization are accounted for even when the supplied beliefs are incorrect. If $k = O(s)$, this gives the same sparsity order as the rate condition for Post-Double Selection. Therefore, the integrated nature of the penalties permits imperfect beliefs, provided the size of the amelioration set remains sufficiently small. The full sampling, approximation, penalty, design, and rate conditions are given in Appendix A.1.

Under these conditions, the final OLS regression can be used to estimate the treatment effect and its standard error. Let $\hat{u}_i = y_i - \hat{\alpha}d_i - x_i'\hat{\beta}$ denote the residuals from (3.13), and let $\hat{v}_i$ denote the residuals from regressing d_i on the same selected controls. For $J_n = \mathbb{E}[v_i^2]$ and $H_n = \mathbb{E}[v_i^2 u_i^2]$, the asymptotic variance and its estimator are given by

$$V_n = \frac{H_n}{J_n^2}, \quad \hat{V}_n = \frac{\mathbb{E}_n[\hat{v}_i^2 \hat{u}_i^2]}{(\mathbb{E}_n[\hat{v}_i^2])^2}. \quad (3.30)$$

The resulting heteroscedasticity-robust standard error is $\sqrt{\hat{V}_n/n}$. The following theorem establishes its validity for Post-Double Selection with the amelioration set penalties.

Theorem 1 (Inference with amelioration set penalties). *Suppose Assumptions 1–5 in Appendix A.1 hold and the estimator is defined by (3.27) and (3.13), with weights satisfying (3.26). The final OLS is identified with probability tending to one, and*

$$\sqrt{n}(\hat{\alpha} - \alpha_0) = J_n^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n v_i u_i + o_{\mathbb{P}}(1). \quad (3.31)$$

Moreover, $\hat{V}_n - V_n \rightarrow_{\mathbb{P}} 0$ and

$$\frac{\sqrt{n}(\hat{\alpha} - \alpha_0)}{\sqrt{\hat{V}_n}} \xrightarrow{d} N(0, 1). \quad (3.32)$$

Consequently, for fixed $0 < \xi < 1$, the interval

$$\left[\hat{\alpha} \pm \Phi^{-1}(1 - \xi/2) \sqrt{\hat{V}_n/n} \right] \quad (3.33)$$

has asymptotic coverage $1 - \xi$, where Φ is the standard normal distribution function. If the constants, moment bounds, rates, and event probabilities in the assumptions hold uniformly over

classes $\mathcal{P}_n$, coverage is uniform:

$$\sup_{P \in \mathcal{P}_n} |P\{\alpha_0(P) \text{ belongs to (3.33)}\} - (1 - \xi)| \longrightarrow 0. \quad (3.34)$$

The proof follows the Post-Double Selection argument of Belloni et al. (2014a), with an adjustment to accommodate the lower penalization of the amelioration set. More specifically, the controls in A are included in the support used to bound the preliminary estimation errors, such that the size of this support is bounded by $s + k$. Accounting for the controls in A therefore allows the baseline penalty to control the remaining regressors, even when some controls in the amelioration set receive no penalization. The first- and second-stage selections then provide the approximations needed for valid inference on α_0 . The full derivation is given in Appendix A.

The result accommodates the Discrete and Half-Discrete penalties, as well as data-dependent weights that satisfy (3.26) and the conditions of the minimisation problem. Therefore, valid asymptotic inference does not require perfect confounding variable selection or correct beliefs about every control in the amelioration set. However, the extent to which lower penalization improves bias and coverage in the finite sample depends on the information contained in the beliefs and the relationships between the controls. The Monte Carlo simulations will explore these differences across the designs and error distributions introduced in the methodology.

4 Data

Monte Carlo simulations will accommodate the exploration of the amelioration set penalties under different conditions, such as varying error distributions, different first- and second-stage R^2 values, and different DGPs. A complete description of the testing conditions is given in Section 5. In addition, the findings of the theoretical exploration are applied in a re-examination of an empirical case covered in Belloni et al. (2014a), specifically the effect of abortion on crime rates (Donohue & Levitt, 2001), to uncover the impact of the amelioration set penalties on the estimated treatment effect and any differences in variable selection and the economic interpretations of coefficient estimates.

In their research, Donohue and Levitt (2001) evaluate the causal impact of abortion rates on three different types of crime: property, violence, and murder. The original set of controls includes lagged prisoners per capita, lagged police per capita, unemployment rate, per-capita income, poverty rate, welfare generosity, concealed weapons laws, and beer consumption per capita. Belloni et al. (2014b) find that estimating the causal impact is challenging due to non-random state-level abortion rates and the potential influence of other confounding factors related to both abortion and crime rates, providing a strong motivation for the use of Post Double Selection. They expand the set by including state-specific effects, time-specific effects, a set of control variables to account for time-varying state-level factors, and a variety of complex transformations, yielding the following model:

$$y_{cit} - y_{cit-1} = \alpha_c(a_{cit} - a_{cit-1}) + z'_{cit}k_c + g_{ct} + \eta_{cit}, \quad (4.1)$$

where y_{cit} is the crime rate and a_{cit} is the abortion rate for crime type $c \in \{\text{property, violent,}$

murder}, α_c is the treatment effect of abortion on crime, g_{ct} represents time effects, η_{cit} is the error term, and z_{cit} is an enriched set of the original controls, including higher-order terms, interactions with controls, a quadratic time trend, initial level differences, and within-state differences.⁴ The z_{cit} term can be considered a high-dimensional control set that can be reduced to an ASM, for which it is assumed that an approximation using only a small number of controls yields a small enough approximation error.

The resulting data includes the same state-level information as reported by Donohue and Levitt (2001), excluding Alaska, Hawaii, and Washington, D.C., resulting in a sample of 48 cross-sectional observations on yearly state-level data from 1985 to 1997 (12 time series observations), yielding a total of 576 observations. The control set comprises 284 variables for each of the three types of crimes. This implies that for determining the state-specific effect, there are $n = 12$ observations for $p = 284$ controls, making it a case where $n \ll p$, and hence a good opportunity to evaluate the performance of Ω . In this empirical context, the amelioration set has been previously defined in the literature as the original set of 7 controls (Belloni et al., 2014a). Therefore, this same set will be used in this empirical re-examination. Given the economically intuitive nature of the amelioration set, its inclusion in the Post Double Selection will yield more straightforward and meaningful economic interpretations compared to more complex terms.

5 Methodology

In this section, the setup of the Monte Carlo simulations under different data properties is described, the varying designs of Data Generation Processes are introduced, the necessary procedures for a valid Post Double Selection are outlined, different data-independent and data-dependent amelioration set penalties are introduced, and the metrics for evaluating their performance are presented.

5.1 Monte Carlo Simulations

Given the theoretical focus of this paper and the difficulty in measuring the specific effect of the amelioration set penalties on the complex interactions within the control selection, Monte Carlo simulations are used. Each estimation is run for 1,000 simulations to ensure a sufficiently large sample for valid Monte Carlo properties.

5.1.1 DGP Setup

Following Belloni, Chernozhukov and Hansen (2013), the structure of the DGP is based on the partially linear models (3.3) and (3.4):

$$y_i = d_i' \alpha_0 + x_i' \beta_0 + u_i, \quad u_i \sim \mathcal{N}(0, \sigma_u^2) \quad (5.1)$$

$$d_i = x_i' \eta_0 + v_i, \quad v_i \sim \mathcal{N}(0, \sigma_v^2) \quad (5.2)$$

for a design matrix $X = (x_1, \dots, x_n)'$ containing $p = 200$ regressors across $n = 100$ observations, $x \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{kj} = (0.5)^{|j-k|}$ for $j, k \in \{1, \dots, p\}$, and the treatment effect $\alpha_0 = 1$. The

⁴The full set of variables and transformations can be found in Belloni, Chernozhukov and Hansen (2013).

design matrix is normalised such that $\mathbb{E}_n[x_{ij}^2] = 1$ for $j \in \{1, \dots, p\}$, placing the controls on the same empirical scale for unit penalty loadings under homoscedastic errors. Each simulation draws new x_i 's, u_i 's, and v_i 's from their respective distributions.

By construction, this partially linear model cannot be estimated using OLS, making it an ideal testing ground for evaluating the sparsity approximation abilities of Post-Double Selection and its extensions in finite samples. Three distinct DGP designs are considered to evaluate control selection in finite samples.⁵ Design 1 is a Linear decay:

$$\beta_0 = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 0, 0, 0, 0, 0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 0, \dots, 0\right)', \quad (5.3)$$

$$\eta_0 = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, \frac{1}{9}, \frac{1}{10}, 0, \dots, 0\right). \quad (5.4)$$

Design 2 is a Quadratic decay:

$$\beta_0 = \left(\frac{1}{1}, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, 0, 0, 0, 0, 0, 1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, 0, \dots, 0\right)', \quad (5.5)$$

$$\eta_0 = \left(1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \frac{1}{6^2}, \frac{1}{7^2}, \frac{1}{8^2}, \frac{1}{9^2}, \frac{1}{10^2}, 0, \dots, 0\right). \quad (5.6)$$

and Design 3 is an extended Quadratic decay:

$$\beta_0 = \left(\frac{1}{1}, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, 0, 0, 0, 0, 0, 1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \dots, \frac{1}{200^2}\right)', \quad (5.7)$$

$$\eta_0 = \left(1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \frac{1}{6^2}, \frac{1}{7^2}, \frac{1}{8^2}, \frac{1}{9^2}, \frac{1}{10^2}, \dots, \frac{1}{200^2}\right). \quad (5.8)$$

The three designs are chosen for their different extents of approximating a sparse representation. Design 1 has the clearest representation, with a sparse design that uses sizable coefficients in s . Design 2 has a similarly sparse representation but with significantly lower coefficients in s . Design 3 deviates slightly from Design 2, including continuously decreasing, yet possibly significant, coefficients throughout the full structural representation. The coefficient vector in Design 3 is dense. Whether it is approximately sparse depends on how accurately a small subset of controls approximates the full regression function. Due to their structures, it is expected that in Design 1, the vanilla Post-Double Selection will capture the exact number of confounding variables; in Design 2, the estimator will capture only some of the confounding variables; and in Design 3, the estimator will capture too many. Therefore, the amelioration set penalties are expected to produce the greatest gains in the latter two designs.

5.1.2 Error Distribution

The amelioration set penalties are evaluated under both homoscedastic and heteroscedastic error terms, as in practice, perfect homoscedasticity is not assumed. Under the assumption of homoscedasticity, $\sigma_u = \sigma_v = 1$ and $\sigma_{uv} = 0$. Under the assumption of heteroscedasticity, the

⁵The DGP designs are inspired by a combination of those used in Belloni and Chernozhukov (2011) and Belloni et al. (2014a).

following specification is considered (Belloni, Chernozhukov & Hansen, 2013):

$$\sigma_d = \sqrt{\frac{(1 + x_i' \beta_0)^2}{\frac{1}{n} \sum_{i=1}^n (1 + x_i' \beta_0)^2}}, \quad \sigma_y = \sqrt{\frac{(1 + \alpha_0 d_i + x_i' \beta_0)^2}{\frac{1}{n} \sum_{i=1}^n (1 + \alpha_0 d_i + x_i' \beta_0)^2}}, \quad (5.9)$$

where the sample averages of $\sigma_d(x_i)^2$ and $\sigma_y(d_i, x_i)^2$ both equal one by construction. This normalisation keeps the average conditional error variance comparable to the homoscedastic case.

5.1.3 Partial R^2 Parameterization

Recalling from the introduction, in a system with significant noise, it seems plausible that the true effects of fundamental variables are masked, and sparsity-inducing methods tend to spuriously choose controls that are primarily correlated with the noise instead of the outcome variable. To evaluate the effectiveness of using amelioration set penalties to mitigate spurious selection, they are tested under varying levels of noise. More specifically, the DGPs of the different designs are scaled such that the ratio of the explainable model variance to the total variance (also called the R^2 of a model) can be specified through a process called ‘*partial R^2 parametrization*’ (Cinelli & Hazlett, 2020). The parameters c_y and c_d are analytically determined for the intended R^2 in both specifications of the partially linear model, such that the following holds:

$$R_d^2 = \frac{\mathbb{V}(\tilde{x}_i' c_d \eta_0)}{\mathbb{V}(\tilde{x}_i' c_d \eta_0 + v_i)} \quad (5.10)$$

$$R_y^2 = \frac{\mathbb{V}(d_i' \alpha_0 + \tilde{x}_i' c_y \beta_0)}{\mathbb{V}(d_i' \alpha_0 + \tilde{x}_i' c_y \beta_0 + u_i)} = \frac{\mathbb{V}((\tilde{x}_i' c_d \eta_0)' \alpha_0 + \tilde{x}_i' c_y \beta_0)}{\mathbb{V}((\tilde{x}_i' c_d \eta_0)' \alpha_0 + \tilde{x}_i' c_y \beta_0 + u_i)}, \quad (5.11)$$

where resulting values of c_y and c_d are specified in Appendix B. The $R^2 \in \{0, 0.2, 0.4, 0.6, 0.8\}$ values do not necessarily have to be equal, allowing for a 3-dimensional grid to show the behaviour of the different amelioration set penalties across varying levels of noise.

5.2 Post-Double Selection

This section describes the practical methodology for practitioners to correctly initialize and make necessary adjustments to the Post-Double Selection method.⁶

5.2.1 Λ -Simulations

The data-dependent λ -penalty introduced in (3.16) relies on the value of $\Lambda(1 - \gamma \mid X)$, where $\Lambda := 2n \|\mathbb{E}_n[x_i g_i]\|_\infty$ is simulated using the design matrix and independent draws $g_i \sim \mathcal{N}(0, 1)$. $\Lambda(1 - \gamma \mid X)$ is the conditional $(1 - \gamma)$ -quantile of these simulations, where γ is the tail probability. Under conditionally Gaussian, homoscedastic disturbances and unit loadings, the corresponding

⁶Note that there is already an R package called ‘*hdm*’, written by one of the authors based on the original paper by Belloni et al. (2014a). However, due to its inflexibility in adjusting the penalties within the pre-made function—a requirement for this thesis’ analysis—the code was replicated, with some elements copied and others adjusted (Chernozhukov et al., 2016). See Appendix C for a full description of the various functions used in the code.

penalty is $\lambda = c\hat{\sigma}\Lambda(1-\gamma \mid X)$. The 'hdm' package recommends 5000 simulations, but preliminary tuning showed that 3000 iterations produce negligible differences while significantly reducing computation time.

5.2.2 σ -Estimation

The other important element in determining λ is the estimation of σ . Under conditionally Gaussian, homoscedastic disturbances, the equality $\|\mathbb{E}_n[x_i\varepsilon_i]\|_\infty \stackrel{d}{=} \sigma \|\mathbb{E}_n[x_i g_i]\|_\infty$ holds conditional on X . The difficulty lies in that σ is unobserved; hence, an iterative procedure is used to refine its estimate (Belloni & Chernozhukov, 2011). The Lasso penalty is directly dependent on the value of $\hat{\sigma}$; therefore, each iteration producing a different estimate affects the λ of the next iteration. This continues until successive estimates differ by less than the specified tolerance or the maximum number of iterations is reached. The score condition in (3.23) remains a separate requirement for the inferential result.

The original paper initialises the iterative procedure with a conservative estimate from a regression on a constant. However, this method is adjusted to use a better-informed preliminary OLS regression. For either selection response $q_r \in \{d, y\}$, let $\hat{e}_{r,I_0} = q_r - X_{I_0}\hat{b}_{r,I_0}$ denote its preliminary residual vector. The initial scale estimate is $\hat{\sigma}_{r,I_0} = \sqrt{\hat{e}_{r,I_0}'\hat{e}_{r,I_0}/n}$. The full iterative procedure to estimate $\hat{\sigma}$ using Post-Lasso iterations is given in Appendix D.

5.3 Amelioration Set Penalties

The concept of incorporating beliefs in approximating sparse representations is complex to simulate in Monte Carlo simulations and is underrepresented in the literature. Hence, the following methodology presents an experimental approach to generating 'beliefs' on which controls should be included in the amelioration set. The primary challenge is avoiding data-snooping bias, where one assumes beliefs that should not be known ex-ante. The goal is to specify a sparse DGP with a relatively small number of known confounding variables and many irrelevant controls, assuming approximate sparsity, and to create a sparse approximation based on 'beliefs' that must be specified ex-ante.

Logically, if the parametric model is known, it is easy to provide correct information to the estimator, but this would be an invalid practice. Conversely, purposefully providing the estimator with incorrect information will deliberately make the estimator perform worse than it otherwise would.⁷ Therefore, the resulting methodological setup is experimental. First, the methodology of varying levels of belief accuracy is explained, followed by the different proposed penalization methods.

⁷Consider as an intuitive example trying to determine the treatment effect of an economic policy on inflation. A researcher might have economic intuition about several important confounding controls that should be included and therefore incorporates these beliefs into the Lasso estimations. However, whether these controls are truly confounding will probably be unknown, and therefore, they will not cause data snooping. When one has parametric knowledge, the lower penalization on controls known to be confounding would be considered data-snooping. There is a very fine line to thread in this experimental setup to avoid data snooping.

5.3.1 Generating Ω -beliefs

For generating beliefs, the first $s = 15$ controls form the reference set that the researcher aims to include. The candidate set contains the first k controls, and $\hat{k}$ denotes the number drawn into the realised amelioration set A . The following setup is proposed for $p = 200$ under three levels of precision, reflecting the concentration of reference controls among the candidates:

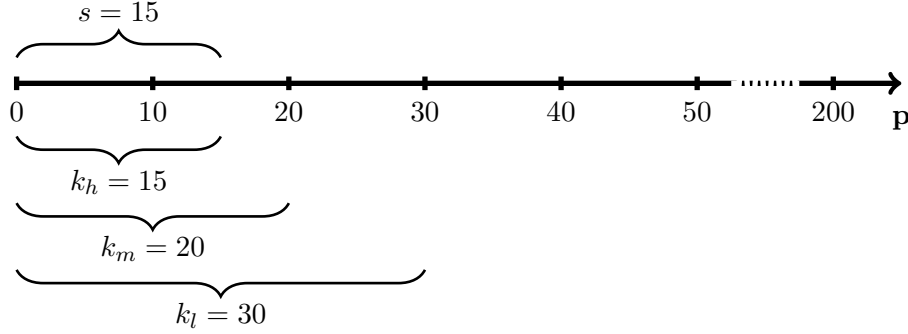


Figure 1. The domain of the amelioration set under varying levels of precision, given $p = 200$

Figure 1 shows the three levels of precision: high (k_h), medium (k_m), and low (k_l), in comparison to s . A lower level of precision enlarges the candidate set beyond the reference controls. Each reference control has the same probability of inclusion across the three levels, while a larger candidate set increases the expected number of other controls included. This approach reflects the practitioners' methodology: having focused knowledge about which factors drive the DGP versus casting a wider net when such knowledge is lacking.

The realised amelioration set A is drawn from the first k_i candidate controls for each precision level $i \in \{h, m, l\}$. Each candidate is included independently with probability 0.5, so $\hat{k} = |A| \leq k_i$ and $\mathbb{E}[|A|] = k_i/2$. This allows the set of beliefs to vary across simulations, although drawing the entire candidate set remains possible. With $s = 15$ reference controls, the probability that a uniformly drawn candidate is both in this reference set and included is $1/2$, $3/8$, and $1/4$ for the high, medium, and low levels, respectively. Among included controls, the expected share belonging to the reference set, conditional on a nonempty set, is instead 1, $3/4$, and $1/2$.

5.3.2 Data-Independent Penalties

The first choice for Ω is the data-independent '*discrete*' penalty, which depends only on the identities of the controls in A and does not consider other available information in the data. One supporting argument for this penalty is that it counteracts Lasso's potentially spurious reliance on control selection based on correlation with noise. Instead, the penalty is based entirely on the discrete belief regarding whether a control should or should not be included in the amelioration set, guided by economic intuition.

The Discrete penalty, denoted by Ω_D , assigns a penalty of zero if the control is an element of the amelioration set and a penalty of one otherwise. This penalty serves as a baseline comparison to existing literature by integrating the amelioration set within the Lasso estimations. The unpenalized controls are fitted jointly with the other controls and can be profiled out by OLS. Their estimated coefficients can still be zero; guaranteed retention in the final OLS would require

adding them explicitly to the selected set.

A second data-independent penalty is introduced through the ‘*half-discrete*’ penalty, denoted by Ω_{HD} . This penalty functions similarly to Ω_D but assigns a penalty of one-half if the control is an element of A . It serves as a robustness check to determine whether the complete removal of penalization is necessary for the amelioration set or if a partial penalty, such as one-half, yields similar results. Exploring the effects of scaling the Discrete penalty to other values within its domain is an interesting idea for future research.

$$\omega_{D,rj} = \begin{cases} 0, & j \in A, \\ 1, & j \notin A, \end{cases} \quad \omega_{HD,rj} = \begin{cases} 0.5, & j \in A, \\ 1, & j \notin A. \end{cases} \quad (5.12)$$

5.3.3 Data-Dependent Penalties

Data-dependent penalties are introduced to improve Ω_D by scaling the amelioration set penalties based on the data in the design matrix. The robustness of using these penalties can be argued from two perspectives: On one hand, due to their data dependence, they might succumb to the fallacy of overfitting to noise, similar to the Lasso estimator. On the other hand, selecting a small subset of potentially information-dense controls allows for the exploitation of information that irrelevant controls might have previously masked.

The adaptive penalty, denoted by Ω_A , is based on the logic of the two-step adaptive Lasso (Zou, 2006)⁸. First, an OLS regression is performed on the controls in the amelioration set A . For either selection response $q_r \in \{d, y\}$, the preliminary coefficient vector is

$$b_r = (X_A' X_A)^{-1} X_A' q_r.$$

The multiplicative inverses of these coefficients are then scaled between zero and one using (5.13). This specification uses signed inverse coefficients, such that their signs as well as their sizes determine the resulting weights.

The score penalty, denoted by Ω_S , uses information from the score to set the amelioration set penalties. For a preliminary residual vector $\hat{e}_r$, write $s_r = X_A' \hat{e}_r / n$. The signed scores are scaled between zero and one, such that controls with a smaller signed score receive a lower penalty. Note that the residual vector changes across iterations of estimating $\hat{\sigma}$, so Ω_S is updated between iterations and held fixed within each Lasso minimisation.

$$\text{scale}(\zeta)_j := \frac{\zeta_j - \min_{h \in A} \zeta_h}{\max_{h \in A} \zeta_h - \min_{h \in A} \zeta_h} \quad (5.13)$$

$$\omega_{A,rj} = \begin{cases} \text{scale}((b_{rh}^{-1})_{h \in A})_j, & j \in A, \\ 1, & j \notin A, \end{cases} \quad (5.14)$$

$$\omega_{S,rj} = \begin{cases} \text{scale}(s_r)_j, & j \in A, \\ 1, & j \notin A. \end{cases} \quad (5.15)$$

Here X_A contains the columns indexed by A , and $\Omega_{A,r}$ and $\Omega_{S,r}$ collect the respective weights.

⁸I want to thank Dr. Kaspar Wüthrich for pointing out this idea in an e-mail conversation.

If the preliminary OLS is not uniquely defined, an inverse coefficient is undefined, or all inputs to the scaling rule coincide, the affected weight vector is set to one on A . For an empty A , every weight is one. These conventions keep the weights within the range in (3.26). A full sequential overview of all amelioration set penalties is presented in Appendix E.

5.4 Metrics

The metrics used to evaluate the performance of different estimators and penalties align with existing literature (Belloni & Chernozhukov, 2011; Belloni et al., 2014a). These metrics include the treatment effect’s bias, the bias’s standard deviation, and the rejection probabilities of the 95% confidence intervals. The confidence intervals are calculated using the jackknife standard error estimator, which is preferred over regular heteroscedastic standard error estimators due to its ability to reduce bias, especially in small samples, by systematically leaving out one observation at a time (MacKinnon & White, 1985). This method provides more robust standard errors in the presence of heteroscedasticity and performs strongly under homoscedasticity, accommodating DGPs with either error distribution. In addition, the jackknife method yields more reliable inferences in finite samples, which is crucial in this research where $n = 100$ (Long & Ervin, 2000).

Therefore, in the final OLS regression of the Post-Double Selection, the residuals can be used to consistently estimate the standard error of the treatment effect. The rejection probability should ideally be close to 5%, indicating accurate coverage in the simulated design. Uniform validity is an asymptotic property established under the conditions of Theorem 1. Furthermore, the number of relevant controls in $\hat{I} = \hat{I}_1 \cup \hat{I}_2$, and the total number of chosen controls in $\hat{I}$, are examined to evaluate the effect of the amelioration set penalties on better-performing control selection. Ideally, the selected set should capture the relevant controls while its total size remains low. Exact recovery of the relevant set is not required for the inferential result.

6 Simulation Results

The results of the Monte Carlo simulations are presented in three stages. First, a preliminary analysis of the different Lasso estimators is conducted to holistically evaluate their performance and provide intuition for using both the Post-Double Selection and amelioration set penalties. Second, different cases of R^2 are examined to evaluate the performance of the various amelioration set penalties and levels of precision in specific chosen cases, demonstrating that some penalties perform better under certain conditions. Third, generalized results are presented across the full grid of R^2 values to comprehensively assess the amelioration set penalties. Here, a practitioner’s recommendation is provided for appropriately using the robust amelioration set penalties under varying conditions.

6.1 Lasso-Estimators for Causal Inference

Table 1 displays the results of the different Lasso-estimators performed on Design 1 with homoscedastic errors, and Figure 2 shows the distribution of the estimated treatment effect, of which the shapes align with those found in existing literature (Belloni et al., 2014a).

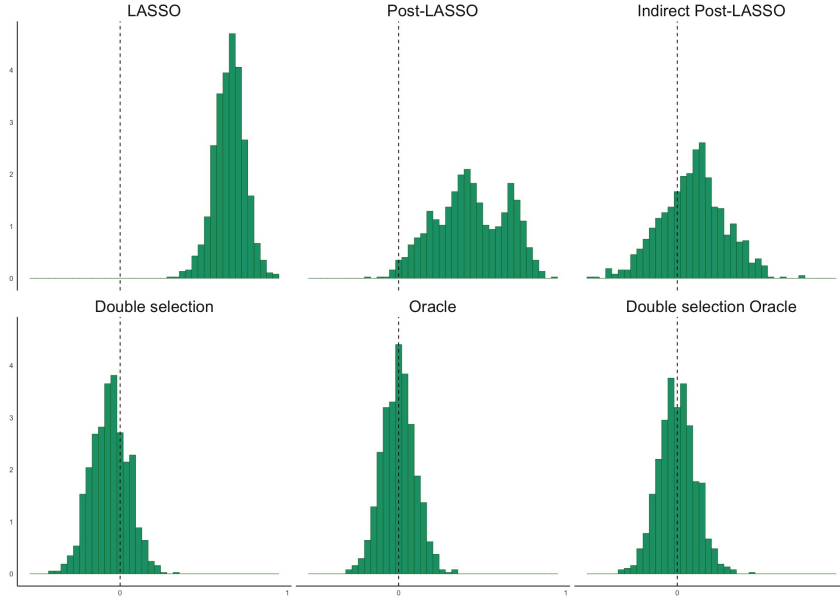


Figure 2. The finite-sample distributions of the different Lasso estimators, described in Section 3.3, are shown. These distributions are based on 1000 simulations of Design 1, as detailed in Section 5.1.1. A dotted vertical line indicates the expected value of the standard normal distribution.

Table 1. Simulation results for Lasso-estimators

Estimator	Mean Bias	Std. Dev	rp(0.05)	Relevant Controls	Total Controls
Lasso	0.653	0.088	1.000	4.8	5.3
Post-Lasso	0.434	0.219	0.879	4.8	5.3
Indirect-Lasso	0.087	0.188	0.006	5.4	5.5
Post-Double Selection	-0.017	0.108	0.097	5.4	5.5
Post-Double Selection Oracle	-0.001	0.110	0.052	15.0	15.0
Oracle	-0.002	0.100	0.055	10.0	10.0

Note: The table reports performance metrics of the different Lasso-estimators, described in Section 3.3, based on 1000 simulations of Design 1 with homoscedastic errors, as detailed in Section 5.1.1, using the metrics detailed in Section 5.4.

The Lasso and Post-Lasso estimators show a significant positive bias. The Lasso bias is expected to result from a combination of ℓ_1 -regularization bias and omitted variable bias (OVB). In contrast, the Post-Lasso corrects for the regularization bias and only suffers from OVB, resulting in its distinct bimodal shape. This result is clarified in Table 1, where, although Lasso and Post-Lasso show similarly low numbers of selected relevant and total controls, the regular Lasso has a higher bias than Post-Lasso. In both estimators, the bias dominates the standard error, leading to a high rejection rate and poor inferential capabilities. Note that this distribution is accentuated in this setup where $n \ll p$ under finite-sample assumptions.

The indirect Post-Lasso, an extension of the Post-Lasso with an additional Lasso selection step of d_i on x_i , shows considerably lower bias and no bimodal shape. The additional step of selecting confounding variables for d_i significantly reduces the OVB, although the standard error remains unaffected. Hence, the magnitude of the test statistic is relatively high, and the rejection probability is the lowest among all estimators and far under the 5% target. This is not a favourable result as it indicates that the standard error bounds are too conservative, leading to type II errors where incorrect inferences are unlikely to be rejected.

The Double-Selection estimator corrects this problem by reducing the variability in the estimated treatment effect through an additional Lasso selection step of y_i on x_i . By doing so, the bias is significantly reduced while also halving the standard error. An intuitive explanation for this trend across all estimators is that each additional control selection step detects more relevant confounding variables in the causality problem by conditioning on the relationships of x_i with both y_i and d_i . This relationship is shown in Table 1, where the number of selected relevant controls monotonically increases. Moreover, there are still gains to be made until the Oracle and Double Selection Oracle are reached. The table shows that a perfect selection of confounding variables leads to negligible levels of OVB and an approximately correct rejection probability of 0.05. This result strongly motivates the use of amelioration set penalties, which could assist in better selecting confounding variables to reach favourable oracle-level performance. However, note that Design 1 with homoscedastic errors is specifically designed to be an approximately sparse model with easy-to-control errors and, therefore, provides optimal conditions for Post-Double Selection.

6.2 ‘Precision’ in the Amelioration Set

The implementation of generating ‘beliefs’ with varying densities of correct information has not resulted in significantly different outcomes across the three levels of precision. Appendix F shows that the differences across precision levels in all metrics are marginal and do not alter the conclusions per precision. The figures do indicate that some of the constructed hypotheses are realized. For instance, as precision increases, the number of selected total controls increases while the number of selected confounding variables converges to s . As a result, the bias decreases slightly, and the coverage is closer to the true value. These graphical results are subtle and difficult to discern. Therefore, ‘medium’ precision is chosen for further analyses as it represents the midpoint and the conclusions drawn are expected to apply to all precision levels.

The most obvious explanation for the methodology’s failure to produce significant results is that different levels of precision only account for varying probabilities of adding more irrelevant controls, whereas the fundamental variables within the DGP are still selected with equal probabilities across all precision levels. Given the strength of Post-Double Selection in selecting correct confounding variables in ASMs, the addition of lower penalizations for irrelevant controls may not have led to significant spurious selection. One possible extension for future research is to calibrate the different precision levels based on both the number of confounding variables and irrelevant controls that are accessible for selection, with higher precision, including a higher ratio of fundamental variables to irrelevant controls.

6.3 Selected R^2 Cases

In this section, the focus shifts to the Post-Double Selection and the amelioration set penalties. Table 2 displays the results of the amelioration set penalties across different designs and error types. The best-performing penalties are identified and contextualized. The Root-Mean-Squared Error (RMSE) is used instead of mean error (bias) to allow negative and positive biases to have an additive effect rather than averaging out.

The specific cases of R^2 values are chosen concerning the existing literature (see Section

2.2), where significantly disparate behaviour is observed between the Post-Lasso and the Post-Double Selection methods based on the different levels of correlation between the confounding variables and the treatment variable, as well as the confounding variables and the outcome. This same analysis is performed. One important difference is that, in the existing literature, the confounding variables in both stages are the same, whereas this is not the case in all current designs. Therefore, these cases provide insight into how the amelioration set penalties, which are purposefully equal in both stages, affect control selection when the correlation is higher in one of the two stages.

Starting with the homoscedastic designs (A, B, and C), a striking observation is that the control estimator performs the worst across all designs and all R^2 cases. The difference is most pronounced in Cases I and III, where the second-stage $R^2 = 0$. This could be because, in the second stage, the relationship between the outcome and the control and treatment variables is difficult to establish, causing the control to struggle in identifying the confounding variables. While some correct confounding variables can be identified in the first stage, the different identities of confounding variables in the equations of the partially linear model leave those that don't overlap unidentified in the second stage.⁹ The amelioration set penalties, which assume some knowledge of the confounding variables in both stages, help mitigate the issue of missing confounding variables in the second stage.

This conclusion aligns with the observation that the Discrete penalty performs the best across all designs. Removing the penalty on the amelioration set allows these controls to enter both selections without a direct penalty. The Half-Discrete penalty, which retains a positive penalty on these controls, produces slightly weaker results. However, these results also highlight a previously identified shortcoming in the methodology: all precision levels assume too perfect knowledge of the real confounding variables. For instance, if the amelioration set does not accurately reflect the DGP and includes a number of irrelevant controls with lower penalties, adding amelioration set penalties might worsen performance compared to the control.

Moreover, under homoscedastic errors, there is a trend indicating that lower absolute penalization of the amelioration set results in better outcomes. However, the question of whether the information in the data-driven penalties is valuable remains open, as scaling from 0 to 1 might not have been sufficient. Perhaps scaling the data-driven penalties from 0 to 0.5, such that the absolute penalization is closer to zero while still exploiting useful information in the data, might produce better results than the Discrete penalty. This sensitivity to scaling is left open for further research.

With heteroscedastic designs (D, E, and F), the performance of different amelioration set penalties becomes more nuanced. The RMSE strictly increases compared to the homoscedastic errors case, which is expected since the final OLS regression under heteroscedastic errors produces estimates with higher variance. While the control estimator consistently shows weak performance across all cases, the previously dominant performance of the Discrete penalty diminishes. Notably, in Case II, the Discrete penalty performs the worst among all estimators in D and nearly as

⁹The difference in coefficient designs between the first and second stages likely explains the difference in results of the Post-Double Selection in Belloni et al. (2014b) across different R^2 cases compared to the results in Table 2.

Table 2. Simulation Results for Selected R^2 Values

Estimation method	<i>Case I</i>		<i>Case II</i>		<i>Case III</i>		<i>Case IV</i>	
	First Stage $R^2 = 0.2$		First Stage $R^2 = 0.2$		First Stage $R^2 = 0.8$		First Stage $R^2 = 0.8$	
	Second Stage $R^2 = 0$		Second Stage $R^2 = 0.8$		Second Stage $R^2 = 0$		Second Stage $R^2 = 0.8$	
	RMSE	Rej. Rate	RMSE	Rej. Rate	RMSE	Rej. Rate	RMSE	Rej. Rate
A. Linear decay with homoscedastic errors								
Control	0.130	0.112	0.128	0.110	0.176	0.108	0.123	0.087
Discrete	0.109	0.064	0.120	0.066	0.133	0.073	0.105	0.046
Half-Discrete	0.118	0.086	0.122	0.086	0.149	0.072	0.115	0.066
Adaptive	0.118	0.082	0.123	0.090	0.151	0.081	0.116	0.069
Score	0.115	0.072	0.122	0.067	0.149	0.100	0.109	0.057
B. Quadratic decay with homoscedastic errors								
Control	0.118	0.090	0.116	0.073	0.168	0.101	0.112	0.071
Discrete	0.112	0.070	0.110	0.053	0.132	0.062	0.109	0.059
Half-Discrete	0.114	0.075	0.113	0.060	0.151	0.063	0.111	0.070
Adaptive	0.114	0.072	0.114	0.061	0.150	0.073	0.110	0.065
Score	0.113	0.075	0.110	0.052	0.150	0.075	0.109	0.066
C. Extended Quadratic decay with homoscedastic errors								
Control	0.116	0.087	0.121	0.091	0.160	0.081	0.112	0.067
Discrete	0.110	0.076	0.112	0.070	0.131	0.058	0.104	0.046
Half-Discrete	0.112	0.066	0.118	0.090	0.145	0.054	0.111	0.057
Adaptive	0.113	0.072	0.118	0.085	0.145	0.063	0.110	0.062
Score	0.112	0.079	0.115	0.070	0.148	0.068	0.106	0.049
D. Linear decay with heteroscedastic errors								
Control	0.186	0.078	0.186	0.077	0.201	0.081	0.176	0.084
Discrete	0.167	0.080	0.201	0.086	0.176	0.087	0.170	0.093
Half-Discrete	0.175	0.080	0.182	0.082	0.187	0.080	0.172	0.088
Adaptive	0.175	0.077	0.189	0.089	0.186	0.084	0.174	0.078
Score	0.170	0.080	0.201	0.088	0.185	0.105	0.168	0.095
E. Quadratic decay with heteroscedastic errors								
Control	0.192	0.098	0.174	0.070	0.200	0.080	0.167	0.074
Discrete	0.180	0.083	0.172	0.068	0.184	0.087	0.164	0.079
Half-Discrete	0.186	0.085	0.169	0.064	0.192	0.072	0.166	0.078
Adaptive	0.187	0.083	0.169	0.061	0.194	0.086	0.164	0.077
Score	0.183	0.083	0.173	0.061	0.186	0.077	0.165	0.075
F. Extended Quadratic decay with heteroscedastic errors								
Control	0.183	0.084	0.185	0.085	0.192	0.079	0.173	0.077
Discrete	0.170	0.079	0.184	0.076	0.178	0.076	0.168	0.081
Half-Discrete	0.175	0.070	0.181	0.078	0.187	0.067	0.172	0.087
Adaptive	0.176	0.072	0.180	0.072	0.189	0.077	0.173	0.078
Score	0.172	0.072	0.185	0.075	0.187	0.083	0.168	0.072

Note: The table reports the RMSE and the rejection probabilities of 95% confidence intervals for the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties, as described in Section 5.3, for selected sets of first-stage and second-stage R^2 . A ‘medium’ precision level is assumed for the amelioration set penalties. The results are based on 1,000 simulations of Designs 1, 2, and 3 with homoscedastic and heteroscedastic errors, as detailed in Section 5.1.1, using the metrics outlined in Section 5.4.

poorly as the control in E and F. This is particularly interesting because, under homoscedastic errors, the Discrete penalty consistently performed the best for that case. A possible explanation is that fitting the amelioration set without penalization distorted the selection of other important controls that might have been highly correlated with the amelioration set, leading to their exclusion in the Lasso-estimation. However, this argument is dubious, as it is exactly the opposite of the intended advantage of the amelioration set penalties.

An important observation is that the Discrete penalty continues to perform well under a second-stage $R^2 = 0$. This makes sense, as in situations with low correlation, the exact structure of the confounding variables is unknown, making any correct information on their identities through the amelioration set valuable. This finding is consistent with literature suggesting that the Post-Double Selection method works best when the second-stage R^2 is low and the first-stage R^2 is high (Belloni et al., 2014b). Therefore, the Discrete penalty shows promise for its integration when precise beliefs about the amelioration set exist. For this reason, the following section will specifically focus on the behaviour of the Control versus the Discrete penalty.

6.4 The Discrete penalty

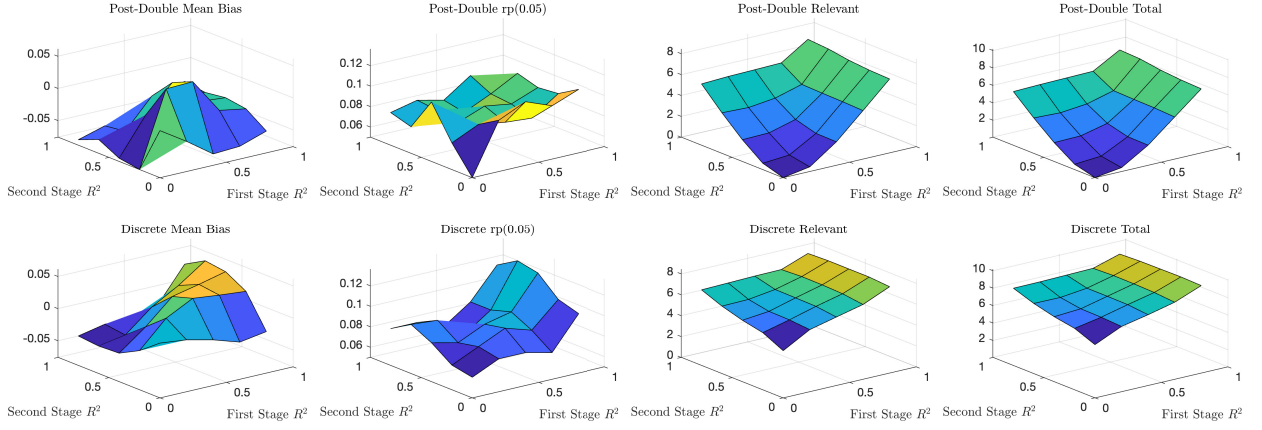
The trends in design, bias and rejection probability are difficult to establish through Table 2 as both the design and the rejection probabilities do not follow a specific pattern given the selected cases. Therefore, this section introduces a three-dimensional grid in Figure 3 to offer better insight into the strength of the Discrete penalty against the Control under imperfect amelioration set selection. Appendix F displays these grids for all metrics (and all Ω 's) ¹⁰ across all designs and both error specifications. This section will focus only on the heteroscedastic errors to create valid practitioner recommendations primarily aimed at realistic heteroscedastic data. Additionally, the heteroscedastic results have more visible differences in specific metrics features, allowing for a more intuitive graphical analysis. ¹¹

Starting with the Control, the differences between the linear, quadratic, and extended Quadratic decay designs perfectly illustrate how the Post-Double Selection method handles ASMs and sparsely approximates non-ASMs for a low enough approximation error. The difference between the Linear and Quadratic decay designs is expected, as the coefficients in the latter drop much more quickly. Therefore, some controls need not be selected for a valid approximately sparse approximation with a sufficiently low approximation error. Under the assumption that not selecting controls with coefficients below a certain threshold in the true DGP does not significantly alter the approximation error, the sparse approximation of the Linear decay will have to select more controls for a similar approximation.

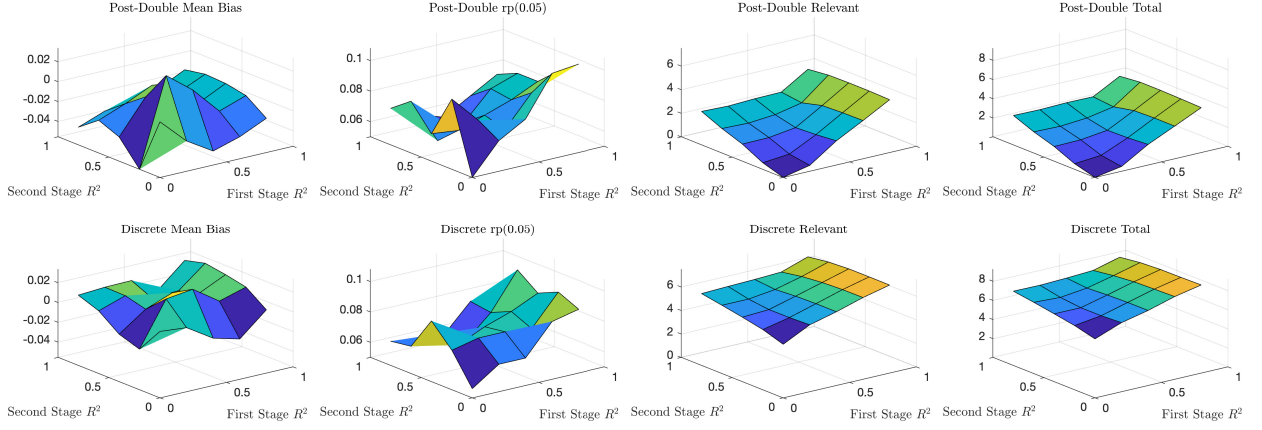
This leads to several outcomes. First, under unfavourable conditions where correct variable selection is challenging (such as low R^2 in either stage), the Linear decay displays higher bias. This can be attributed to the larger coefficient sizes in the Linear decay, meaning that if a confounding variable is not selected—a likely scenario in this finite sample setup—the resulting

¹⁰Further analysis of the other amelioration set penalties is an interesting extension to uncover the effects of data-driven versus data-independent penalties. However, considering the scope of this thesis, only the Discrete penalty can feasibly be evaluated in detail.

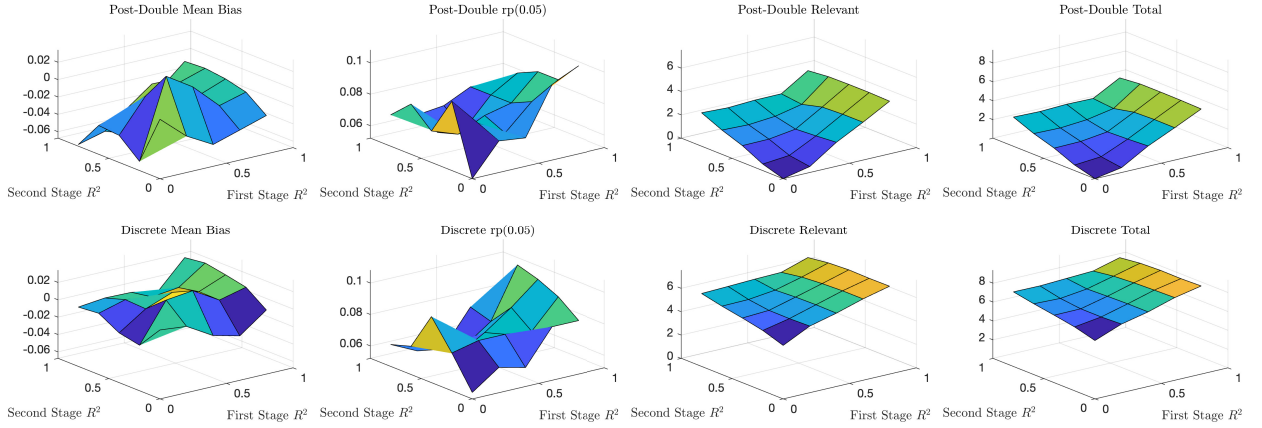
¹¹The values of the bias and rejection probability in the heteroscedastic designs have significantly more mountainous, less uniform shapes. This accommodates a more graphically intuitive comparison across the designs.



(a) Design 1 (Linear decay) with Heteroscedastic Errors



(b) Design 2 (Quadratic decay) with Heteroscedastic Errors



(c) Design 3 (extended Quadratic decay) with Heteroscedastic Errors

Figure 3. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (Control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of Designs 1, 2, and 3, which include heteroscedastic errors as detailed in Section 5.1.1.

OVB will be more significant. This is especially important given the imperfect selection of the full set of confounding variables in the amelioration set for a given level of ‘precision’. Second, the larger coefficient sizes in the true DGP of the Linear decay increase the number of significantly large coefficients that need to be included in the ASM, reflected in the number of selected relevant and total controls. Furthermore, the rejection probability of the quadratic designs appears to be closer to the true value, indicating that the behaviour of the Control is more valid for quadratic designs.

The trend is clear: Quadratic decay leads to a sparser approximation, making it less prone to missing the selection of confounding variables with significant effects in the DGP. This conclusion also explains the similar behaviour between Quadratic decay and extended Quadratic decay. It was unexpected that the extended Quadratic decay, which is not sparse by design, is approximated just as well as a regular Quadratic decay. This is likely due to the threshold idea, where a control with sufficiently small significance in the true DGP is ignored because of its minimal impact on the approximation error. Considering that some controls were already ignored in the Quadratic decay, it is not implausible that all other controls with even smaller effects are similarly ignored.

The implementation of the Discrete penalty has asymmetrical effects on the Linear and Quadratic decay designs. The bias flattens and converges around zero in the Quadratic decay designs across all values R^2 . One explanation is that for all values of R^2 , the Discrete penalty increases relevant and total control selection in these simulations by removing the Lasso penalty on the amelioration set, which is reflected in the flat near-uniform surface at a high base-line level in Figure 3. Increased (correct) control selection could have led to a reduced level of OVB.

The most interesting finding is the bias of the Discrete penalty on the Linear decay, which seems to show little improvement over the control in terms of absolute bias. The bias stays at a near-zero level for moderate levels of first-stage and second-stage R^2 . However, especially for a high value of second-stage R^2 , the bias is volatile to the change in first-stage R^2 . This behaviour is more moderate given a lower level of first-stage R^2 . The surface strongly resembles that of the Post-Lasso in a study in the literature conducted on a DGP variant of the extended Quadratic decay by Belloni et al. (2014a), which has also been described in Section 2.2. In that case, the Post-Lasso had trouble under a high first-stage R^2 to select all relevant confounding controls in the second-stage estimation. This could be the same here, where the Discrete penalties, which are equal for the first-stage and second-stage equations, hinder the selection of the true confounding variables by leaving irrelevant controls unpenalized. Given the higher coefficient values of those controls in the Linear decay and the impact of missing the selection of a relevant control on the bias, this could be the reason for the turbulent bias plot.

Therefore, based on the analysis of the Monte Carlo simulations, the following recommendation for practitioners is offered: data-independent Discrete penalties can yield significant improvements across different ASMs, with performance varying according to the first and second-stage correlations. The optimal results are achieved with a moderate to high first-stage R^2 and a low second-stage R^2 . Under favorable conditions, the bias is significantly reduced across all correlations, while under unfavorable conditions, it does not worsen compared to the control.

7 Empirical Case: The Effect of Abortion Rate on Crime

Building on the practitioner’s recommendation, the empirical case of Donohue and Levitt (2001) is revisited to evaluate the effect of the amelioration set penalties, particularly the Discrete penalty, on the treatment effect, control selection, and the economic interpretations of the economically intuitive controls. Table 3 presents the different estimations of the treatment effect, while Table 4 shows the coefficient estimates of the economically intuitive variables. These are defined as the set of controls used in the original research.¹² In the appendix, Table G.1 shows the cardinalities of selected relevant and total controls, and Tables G.2 to G.4 show the effect of amelioration set penalties on control selection.

To improve readability, the terms will be referred to as follows: Post-Double Selection will be called the ‘Control’, Post-Double Selection with the ex-post addition of the amelioration set will be called the ‘Amel-Control’, and Post-Double Selection with the Discrete penalty for the amelioration set will be called the ‘Discrete’.

The R^2 values of the different first-stage and second-stage regressions across crime types—violent crime, property crime, and murder—have been established in the literature (Belloni et al., 2014b). Respectively, the first-stage R^2 values are 0.8420, 0.6116, and 0.7781, while the second-stage R^2 values are 0.0251, 0.1179, and 0.0039. Based on the practitioner’s recommendation, these values correspond to regions of the R^2 grid where the Discrete can be a valuable addition to the Control, assuming some degree of precision in identifying the amelioration set. In the empirical case, it is therefore assumed that the set of economically intuitive variables includes some significant information on the true confounding variables.

Table 3. Estimated treatment effect of abortion on different crimes

	Violent Crime		Property Crime		Murder	
	Effect	Std. Error	Effect	Std. Error	Effect	SE
Donohue and Levitt (2001)	-0.129	0.024	-0.091	0.018	-0.121	0.047
Belloni et al. (2014b)	-0.104	0.107	-0.030	0.055	-0.125	0.151
Belloni et al. (2014b) + ex-post Amel.	-0.082	0.106	-0.031	0.057	-0.068	0.200
Control	-0.158	0.120	-0.024	0.043	-0.117	0.417
Control + ex-post Amel.	-0.133	0.122	-0.028	0.045	-0.103	0.434
Discrete	-0.113	0.120	-0.013	0.044	-0.063	0.421

Note: The table reports the estimated treatment effects of abortion on three types of crimes in the empirical study, including results in literature, for the Post-Double selection, and the Post-Double Selection with the ex-post addition of the amelioration (Amel.) set, under heteroscedastic errors. Standard error estimates are produced using the jackknife method.

The replication results align with those found in the literature, showing similar values for the Control and Amel-Control. As in previous studies, the estimated treatment effects from the original research are significantly different for property crime and slightly for violent crime. All standard errors for the Control and its extensions are large, with 95% confidence intervals spanning a broad positive and negative range. Therefore, drawing valid economic conclusions from these results with sufficient certainty is challenging.

¹²This follows the methodology used in previous re-examinations of Donohue and Levitt (2001) using the Post-Double Selection (Belloni et al., 2014b)

Table G.1 reveals an important observation: none of the Control estimations selects economically intuitive variables. Instead, more complex interactions and time-dependent controls are chosen. For property crime, the Discrete chooses one less variable than the Control but does choose five economically intuitive variables. For murder, the Discrete selects four additional controls, all of which are economically intuitive. Due to the disparity in control selection, the Discrete shows a larger deviation from the Control, with the treatment effect being approximately halved for both property crime and murder. However, since the true effect is unknown, it is difficult to assess this result and conclude whether a higher number of economically intuitive penalties is favourable.

Another interesting observation is that the treatment variable estimates of the Amel-Control surprisingly fall between those of the Control and the Discrete. Therefore, the Discrete penalty, which should be a less extreme and more robust alternative to the Amel-Control, selects controls considerably differently. To uncover the true treatment effect, the treatment variable should be assumed to be exogenous within the large control set, which could be better accommodated through the alternative selection method of the Discrete penalties. Thus, examining the selected variables across different methods may provide better insights into the correct confounding variable selection.

Several trends on control selection emerge across Tables G.2 to G.4. First, as a robustness check, the variable selection between Belloni et al. (2014b) and the Control is largely similar. The Control generally includes all the controls used in the literature and adds a few extra ones, mostly interactions with time, which already comprise most of the variable selection. Second, as expected, the variables selection between the Amel-Control and the Discrete is very similar. For example, only one variable was selected out of 18 for violent crimes. However, the treatment effect estimates differ significantly, ranging from -0.133 to -0.113. Similarly, for murder, the Control selects three additional controls for a total of 15, while the ex-post amelioration set halves the estimated effect from -0.103 to -0.063. From here, the question arises as to how such a minor difference in Control selection can significantly affect the estimated effect and whether the controls excluded by the integration of the Discrete undermine the exogeneity assumptions of the treatment variable. The disparity in control selection is most pronounced for property crime. The Amel-Control set selects 24 controls, while the Discrete penalty selects a subset of 16 of those controls. Of the eight missing controls, only two are economically intuitive, illustrating the Discrete’s tendency to prioritize economically intuitive variables over others.

The coefficient estimates of the selected controls presented in Table 4 allow for a final economic interpretation. Generally, the differences in coefficient estimates are slight, with the maximum deviation being by a factor of two. The only outlier is ‘*Dinc*’, which represents the change in income. This variable shows a significantly higher estimate but is not selected in two of the Discrete estimations. A zero penalty does not guarantee a nonzero estimated coefficient. This non-selection alone is therefore insufficient to establish that the variable has little confounding relevance. The table also displays changes in sign for variables such as ‘*Dbeer*’ and ‘*Dpov*’ in the murder estimation. Given that ‘*Dinc*’ was forced into the ex-post amelioration set estimation, its non-selection in the Discrete penalty naturally creates a different interpretation of the economically intuitive variables. Whether this difference in interpretation more closely

Table 4. Coefficient estimates of economically intuitive variables

	Violent Crime		Property Crime		Murder	
	Ex-Post	Discrete	Ex-Post	Discrete	Ex-Post	Discrete
D (treatment effect)	-0.133	-0.114	-0.028	-0.013	-0.103	-0.063
Dinc	16.4	23.1	-4.74	-	-230	-
Dpov	0.161	0.152	0.069	0.092	-0.112	0.064
Dafdc	0.034	0.040	-0.016	-0.020	-0.362	-0.368
Dbeer	0.086	0.164	-0.047	-0.099	0.267	-0.109
Dpolice	-0.019	-0.017	-0.026	-0.031	0.337	0.334
Dprison	-0.048	-	-0.019	-0.019	-0.020	-

Note: The table reports the coefficient estimates of the economically intuitive variables in the Post-Double Selection with the ex-post addition of the amelioration set and the Post-Double Selection with the Discrete penalty. A '-' indicates that the variable was not selected.

resembles reality remains unanswered.

8 Conclusion

The Post-Double Selection method effectively reduces bias in causal inference on treatment effects, particularly in settings with high-dimensional controls. Selecting spurious confounding variables can lead to significant bias in noisy conditions and finite samples. This thesis proposes a methodology to mitigate this bias by conditioning on beliefs about the true structural representation through a targeted set of regressors, known as the amelioration set. Various data-dependent and data-independent penalties are evaluated for their effectiveness in reducing bias and improving inference. The Discrete penalty, a robust adaptation of current literature that integrates information on the amelioration set into the Post-Double Selection methodology, has shown potential for enhancing inference across various designs, error distributions, and correlations between the outcome, treatment, and controls. This finding assumes sufficient knowledge of economically intuitive variables driving the DGP. When applied to empirical situations, the Discrete penalty functions as a hybrid of the Post-Double Selection and an extended version with an ex-post amelioration set addition.

Acknowledging the limitations of this research, the Monte Carlo simulation results may rely too heavily on accurate knowledge of the amelioration set, potentially overstating results through data-snooping. Therefore, analysing scenarios with incorrect information would be valuable to determine the robustness of the Post-Double Selection method when faced with a poorly chosen amelioration set. The treatment of error distributions has also been polarised: simulations either assumed homoscedasticity with appropriate equivariance correction or heteroscedasticity with the same correction. However, the scenario in which the wrong type of correction is applied has not been considered. This is particularly relevant because, in practice, the error distribution is very sensitive and difficult to determine when using such a complex method. In addition, the high rejection probabilities across the simulations suggest that the error might not have been conservative enough. Hence, an idea for future research is to investigate the usage of a fatter-tailed distribution.

The potential of the integrated amelioration set offers an opportunity for its integration

into Bayesian statistics through the Bayesian Lasso. The Discrete penalty has proven strong and is easily implemented by modifying the priors for specific controls. Theorem 1 establishes sufficient conditions under which the integrated amelioration set preserves asymptotically valid inference, including uniform coverage when the conditions hold uniformly. The result permits incorrect beliefs while accounting for their estimation cost through the combined complexity $s+k$. Establishing conditions for a finite-sample improvement over ordinary Post-Double Selection remains a separate theoretical question.

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A Proof of Inferential Validity

The following proof extends the Lasso arguments of Belloni and Chernozhukov (2013) and Bickel et al. (2009) to the amelioration set penalties and applies the resulting bounds to the Post-Double Selection estimator of Belloni et al. (2014a).¹³ The main adjustment is to include all controls receiving lower penalization in the support used for the preliminary estimation bounds. The rate conditions therefore depend on $s + k$, allowing the proof to accommodate both incorrect beliefs and weights equal to zero. The assumptions are stated below, followed by the derivation of Theorem 1 and a penalty calibration satisfying the required score condition.

A.1 Regularity conditions

The following conditions apply to the model and estimator in Section 3.5.3. All quantities may depend on n , although this dependence is suppressed where it does not create ambiguity. Write $\mathbb{E}_n[a_i] = n^{-1} \sum_{i=1}^n a_i$ and $\|a\|_n = \{\mathbb{E}_n[a_i^2]\}^{1/2}$. The constants in the assumptions are fixed and do not depend on n .

Throughout the analysis, measurable minimisers of (3.27) are used, with the weights and loadings held fixed within each minimisation. Undefined data-dependent weight formulas are assigned a measurable value satisfying (3.26); for instance, weight one can be used when a denominator is zero. A fixed number of mandatory controls, including an intercept, can receive weight zero and be retained in the final regression. These controls are included in A and counted in k , and retaining any additional subset of A in the final regression is also allowed.

For the derivation, let $\tau_r = \lambda_r/(2n)$. Dividing the objective in (3.27) by two gives the equivalent criterion $\frac{1}{2}\hat{Q}_r(b) + \tau_r \sum_j \hat{\psi}_{rj} \omega_{rj} |b_j|$. The following conditions are stated in this normalisation.

Assumption 1 (Sampling and identification). For each n , the observations (y_i, d_i, z_i) are independent and identically distributed and satisfy (3.3) and (3.4), with $\mathbb{E}[|y_i| + |d_i|] < \infty$. The dictionary $x_i = P_n(z_i) \in \mathbb{R}^p$ is a non-random measurable function of z_i . For some $q > 4$ and $0 < c < C < \infty$,

$$\begin{aligned} |\alpha_0| &\leq C, & \mathbb{E}[|u_i|^q + |v_i|^q] &\leq C, \\ J_n := \mathbb{E}[v_i^2] &\geq c, & H_n := \mathbb{E}[v_i^2 u_i^2] &\geq c. \end{aligned} \tag{A.1}$$

The conditional-mean restrictions provide identification, while the lower bound on J_n excludes a treatment perfectly predictable from the controls.

Assumption 2 (Approximate sparsity). There exist deterministic coefficients β_{m0} and β_{g0} such that

$$m(z_i) = x_i' \beta_{m0} + r_{mi}, \quad g(z_i) = x_i' \beta_{g0} + r_{gi}, \tag{A.2}$$

$$|\text{supp}(\beta_{m0}) \cup \text{supp}(\beta_{g0})| \leq s, \quad \|r_m\|_{L_4} + \|r_g\|_{L_4} \leq C\sqrt{s/n}, \tag{A.3}$$

where $s \geq 1$ and $\|r_f\|_{L_4} = \{\mathbb{E}[|r_{fi}|^4]\}^{1/4}$.

¹³Appendix and step references to Belloni et al. (2014a) refer to the author-posted manuscript, available at <https://arxiv.org/pdf/1201.0224v3>.

Here s bounds the union of the two approximating supports, and the fourth-moment restriction is used for variance consistency. The outcome-selection regression consequently satisfies

$$\ell(z_i) = x_i' \beta_{\ell 0} + r_{\ell i}, \quad \beta_{\ell 0} = \alpha_0 \beta_{m 0} + \beta_{g 0}, \quad r_{\ell i} = \alpha_0 r_{m i} + r_{g i}. \quad (\text{A.4})$$

For the proof, define the following complexity and rate sequences:

$$t_n = s + k, \quad L_n = \log(p \vee n \vee 3), \quad b_n = \sqrt{L_n/n}, \quad a_n = \sqrt{t_n} b_n, \quad (\text{A.5})$$

where $\vee$ denotes the maximum. The following conditions concern the baseline penalty, the design matrix, and the growth of $s + k$, respectively.

Assumption 3 (Baseline penalties). For fixed $0 < \vartheta < 1$ and positive constants $c_\psi, C_\psi, c_\tau, C_\tau$, the following inequalities hold simultaneously for $r \in \{m, \ell\}$ and $j \leq p$, with probability tending to one:

$$c_\psi \leq \hat{\psi}_{rj} \leq C_\psi, \quad c_\tau b_n \leq \tau_r \leq C_\tau b_n, \quad (\text{A.6})$$

$$|\mathbb{E}_n[x_{ij} e_{ri}]| \leq \vartheta \tau_r \hat{\psi}_{rj}. \quad (\text{A.7})$$

This condition requires the baseline penalties to dominate the noise scores. It does not require the reduced penalties $\hat{\psi}_{rj} \omega_{rj}$ to dominate those scores, which would exclude the Discrete penalty. Appendix A.3 gives one explicit calibration satisfying the condition under an additional tail restriction.

Assumption 4 (Design regularity). For fixed positive constants $\underline{\phi}, \bar{\phi}$, and C_x , and a deterministic sequence $h_n \rightarrow \infty$, the following inequalities hold with probability tending to one:

$$\underline{\phi} \|b\|_2^2 \leq \|Xb\|_n^2 \leq \bar{\phi} \|b\|_2^2 \quad \text{for all } \|b\|_0 \leq \min\{p, \lceil h_n t_n \rceil\}, \quad (\text{A.8})$$

$$\max_{i \leq n, j \leq p} |x_{ij}| \leq C_x \sqrt{L_n}. \quad (\text{A.9})$$

These restrictions control collinearity within small collections of controls and the size of individual regressors. An example is given in Appendix A.3; more elaborate transformations of the controls require verification of the same conditions.

Assumption 5 (Growth). The sparsity and amelioration-set bounds satisfy

$$\frac{(s + k) \log(p \vee n \vee 3)}{\sqrt{n}} \rightarrow 0. \quad (\text{A.10})$$

The growth condition counts all controls in the amelioration set, including those that do not contribute to the sparse approximation.

A.2 Proof of Theorem 1

For an index set I with full-column-rank X_I , write

$$P_I = X_I (X_I' X_I)^{-1} X_I', \quad M_I = I_n - P_I. \quad (\text{A.11})$$

For the empty set, $P_\emptyset = 0$. For $a \in \mathbb{R}^n$, use $\|a\|_{n,r} = (n^{-1} \sum_i |a_i|^r)^{1/r}$, with $\|a\|_n = \|a\|_{n,2}$. The final residualised treatment is $\hat{v} = M_{\hat{I}}d$. The proof will first establish

$$|\hat{I}| = O_{\mathbb{P}}(t_n), \quad \|M_{\hat{I}}m\|_n + \|M_{\hat{I}}g\|_n = O_{\mathbb{P}}(a_n), \quad (\text{A.12})$$

and then use these bounds to prove the inferential result.

All probability statements below can be interpreted uniformly when the assumptions hold uniformly. The notation C denotes a finite constant that may change between lines but does not depend on n , p , or the data-generating process within the specified class. Arguments are made on the events in Assumptions 3 and 4; their complements have vanishing probability. On exceptional events where an OLS inverse does not exist, define the estimator by any fixed convention. This does not affect the conclusions.

A.2.1 A consequence of the sparse-eigenvalue condition

The sparse-eigenvalue condition gives the restricted-eigenvalue bound used in the Lasso argument. Following the block decomposition in Bickel et al. (2009, Appendix A, proof of Lemma 4.1(ii)), set

$$c_0 = \frac{1 + \vartheta}{1 - \vartheta} \frac{C_\psi}{c_\psi}, \quad \kappa = \sqrt{\phi}/2. \quad (\text{A.13})$$

For sufficiently large n , condition (A.8) implies, simultaneously for $|T| \leq t_n$ and $\|b_{T^c}\|_1 \leq c_0 \|b_T\|_1$,

$$\|Xb\|_n \geq \kappa \|b_T\|_2. \quad (\text{A.14})$$

For completeness, choose a fixed $D \geq \max\{1, 4c_0^2\bar{\phi}/\phi\}$ and partition T^c into blocks $B_1, B_2, \dots$ of size $\lceil Dt_n \rceil$, ordered by decreasing $|b_j|$. The last block may be smaller. The ordering gives $\sum_{j \geq 2} \|b_{B_j}\|_2 \leq c_0 \|b_T\|_2 / \sqrt{D}$. Since $h_n \rightarrow \infty$, (A.8) eventually applies to $T \cup B_1$ and each block, so the triangle inequality gives

$$\|Xb\|_n \geq \sqrt{\phi} \|b_{T \cup B_1}\|_2 - \sqrt{\phi} \sum_{j \geq 2} \|b_{B_j}\|_2 \geq \left(\sqrt{\phi} - c_0 \sqrt{\phi/D} \right) \|b_T\|_2 \geq \kappa \|b_T\|_2.$$

If fewer coordinates remain, take them all in B_1 and the tail sum is zero; when $T^c = \emptyset$, take $B_1 = \emptyset$.

A.2.2 A weighted Lasso prediction bound

The prediction bound is obtained by adapting the basic-inequality argument in Bickel et al. (2009, Lemma B.1) and Belloni and Chernozhukov (2013, Theorem 1). Here the comparison support includes the amelioration set, such that the argument also allows penalties equal to zero.

Lemma A.1 (Enlarging the support). *Consider either preliminary regression, written $q = X\beta_0 + r + e$, and let $\hat{\beta}$ solve (3.27). Set $T = \text{supp}(\beta_0) \cup A$, $\delta = \hat{\beta} - \beta_0$, and $c_r = \|r\|_n$. If $|T| \leq t_n$, then*

$$\|X\delta\|_n \leq C\{c_r + \tau\sqrt{t_n}\}. \quad (\text{A.15})$$

Under the stated assumptions, for $r \in \{m, \ell\}$,

$$\left\|X(\widehat{\beta}_r - \beta_{r0})\right\|_n = O_{\mathbb{P}}(a_n), \quad \left\|f_r - X\widehat{\beta}_r\right\|_n = O_{\mathbb{P}}(a_n). \quad (\text{A.16})$$

Proof. Optimality relative to β_0 gives the basic inequality

$$\frac{1}{2} \|X\delta\|_n^2 \leq \frac{e'X\delta + r'X\delta}{n} + \tau \sum_j \widehat{\psi}_j \omega_j (|\beta_{0j}| - |\widehat{\beta}_j|). \quad (\text{A.17})$$

Write $D_T = \sum_{j \in T} \widehat{\psi}_j |\delta_j|$ and $D_{T^c} = \sum_{j \notin T} \widehat{\psi}_j |\delta_j|$. Outside T , $\beta_{0j} = 0$ and $\omega_j = 1$. Inside T , the penalty difference is at most $\widehat{\psi}_j |\delta_j|$, including when $\omega_j = 0$. The score event and Cauchy–Schwarz therefore imply

$$\frac{1}{2} Q^2 + \tau(1 - \vartheta) D_{T^c} \leq c_r Q + \tau(1 + \vartheta) D_T, \quad Q = \|X\delta\|_n. \quad (\text{A.18})$$

If $Q \leq 4c_r$, the claim follows. Otherwise $c_r Q < Q^2/4$, and

$$\frac{1}{4} Q^2 + \tau(1 - \vartheta) D_{T^c} \leq \tau(1 + \vartheta) D_T. \quad (\text{A.19})$$

In particular, $D_{T^c} \leq (1 + \vartheta) D_T / (1 - \vartheta)$, which implies $\|\delta_{T^c}\|_1 \leq c_0 \|\delta_T\|_1$. By (A.14),

$$D_T \leq C_\psi \sqrt{t_n} \|\delta_T\|_2 \leq C_\psi \sqrt{t_n} Q / \kappa. \quad (\text{A.20})$$

Substitution into (A.19) gives $Q \leq 4(1 + \vartheta) C_\psi \tau \sqrt{t_n} / \kappa$. Together with the case $Q \leq 4c_r$, this proves (A.15). Approximate sparsity and Markov’s inequality yield $c_r = O_{\mathbb{P}}(\sqrt{s/n})$. Together with $\tau \asymp b_n$, this proves (A.16). In fact, $c_r / (\tau \sqrt{t_n}) = o_{\mathbb{P}}(1)$ because $s \leq t_n$ and $L_n \rightarrow \infty$. Thus a fixed multiple of $\tau \sqrt{t_n}$ bounds the prediction error with probability tending to one. $\square$

The inequalities hold for every realised sample on the score and design events, simultaneously over all admissible weights in (3.26). Thus estimating those weights using the same sample requires no additional independence argument.

A.2.3 Controlling the number of selected controls

The number of selected controls is bounded through the Karush–Kuhn–Tucker conditions, following Belloni and Chernozhukov (2013, Lemma 2 and its proof, Theorem 3). The argument is applied to selected controls outside A , where the baseline penalty is unchanged, and the remaining controls are counted through k .

Lemma A.2 (Selected-model size). *For each preliminary regression, $|\text{supp}(\widehat{\beta}_r)| = O_{\mathbb{P}}(t_n)$. Hence $|\widehat{I}| = O_{\mathbb{P}}(t_n)$.*

Proof. Again suppress the regression subscript. For every $j \in \text{supp}(\widehat{\beta}) \setminus A$, the Karush–Kuhn–Tucker condition is

$$\mathbb{E}_n[x_{ij}(q_i - x'_i \widehat{\beta})] = \tau \widehat{\psi}_j \text{sign}(\widehat{\beta}_j). \quad (\text{A.21})$$

Writing $q = f + e$ and applying the score event gives

$$|\mathbb{E}_n[x_{ij}(f_i - x'_i \widehat{\beta})]| \geq (1 - \vartheta)\tau c_\psi. \quad (\text{A.22})$$

Let B be any subset of these selected indices with $|B| \leq \lceil h_n t_n \rceil$. By the upper sparse-eigenvalue bound,

$$\begin{aligned} |B|(1 - \vartheta)^2 \tau^2 c_\psi^2 &\leq \left\| X'_B(f - X\widehat{\beta})/n \right\|_2^2 \\ &\leq \overline{\phi} \left\| f - X\widehat{\beta} \right\|_n^2. \end{aligned} \quad (\text{A.23})$$

Lemma A.1 makes the right side at most $C\tau^2 t_n$ with probability tending to one. Choose a fixed M larger than the resulting constant in the bound $|B| \leq Ct_n$. If more than $\lceil Mt_n \rceil$ controls outside A were selected, choose that many as B . For sufficiently large n , $h_n > M + 1$, so this is an admissible subset and contradicts (A.23). At most $k \leq t_n$ further selected controls belong to A . Apply the argument in both regressions and take their union. Adding a subset of A to the final OLS also preserves the bound. $\square$

A.2.4 Approximation and noise projections on the selected model

For the rest of the proof write $I = \widehat{I}$, $P = P_I$, $M = M_I$, and $R_f = Mf$ for $f \in \{m, g, \ell\}$. Since the union contains both preliminary supports, projection optimality and Lemma A.1 imply

$$\|R_m\|_n = O_{\mathbb{P}}(a_n), \quad \|R_\ell\|_n = O_{\mathbb{P}}(a_n), \quad \|R_g\|_n = O_{\mathbb{P}}(a_n), \quad (\text{A.24})$$

where the last conclusion uses $g = \ell - \alpha_0 m$. This proves the approximation claim in (A.12).

The score event also gives

$$\|X'v/n\|_\infty = O_{\mathbb{P}}(b_n), \quad \|X'u/n\|_\infty = O_{\mathbb{P}}(b_n), \quad (\text{A.25})$$

since $u = e_\ell - \alpha_0 v$. Let $T_0 = \text{supp}(\beta_{m0}) \cup \text{supp}(\beta_{g0})$ and $B = I \cup T_0$. Then $|B| = O_{\mathbb{P}}(t_n)$. For $S \in \{I, B\}$, put $G_S = X'_S X_S / n$ and $c_{e,S} = X'_S e / n$, where $e = u, v$. Equations (A.8) and (A.25) imply

$$\|P_S e\|_n^2 = c'_{e,S} G_S^{-1} c_{e,S} \leq \underline{\phi}^{-1} |S| \|X'e/n\|_\infty^2 = O_{\mathbb{P}}(a_n^2). \quad (\text{A.26})$$

For an empty set the projection is zero. These bounds hold for the data-dependent sets because the design and score bounds hold simultaneously over all the required indices and sparse sets. In particular, their sizes are below $h_n t_n$ with probability tending to one.

For $f = m, g$, write

$$R_f = r_f + w_f, \quad w_f = X\beta_{f0} - Pf \in \text{span}(X_B), \quad \|w_f\|_n = O_{\mathbb{P}}(a_n). \quad (\text{A.27})$$

The rate follows from (A.24) and the approximation bound. We also use one elementary in-

equality: for $S \in \{I, B\}$ and any $w = X_S b$,

$$\|w\|_\infty \leq C_x \sqrt{|S|L_n} \|b\|_2 \leq \frac{C_x}{\sqrt{\phi}} \sqrt{|S|L_n} \|w\|_n. \quad (\text{A.28})$$

The first inequality uses (A.9); the second uses (A.8).

A.2.5 Expansion of the treatment-effect estimator

The treatment-effect expansion follows the double-selection argument in Belloni et al. (2014a, Appendix B, proof of Theorem 1, Steps 1–3), using the preliminary bounds derived above for the amelioration set penalties. By Frisch–Waugh–Lovell,

$$\hat{\alpha} - \alpha_0 = \frac{d' M g + d' M u}{d' M d}. \quad (\text{A.29})$$

First, $d = m + v$ and $R_m = M m$ give

$$\frac{d' M d}{n} = \mathbb{E}_n[v_i^2] - \|P v\|_n^2 + \frac{2 R'_m v}{n} + \|R_m\|_n^2 = J_n + o_{\mathbb{P}}(1). \quad (\text{A.30})$$

Here the law of large numbers follows from the moment bound, while the other terms vanish by (A.24), (A.26), and Cauchy–Schwarz. Since $J_n \geq c$, the denominator is bounded away from zero with probability tending to one. The sparse-eigenvalue bound ensures that X_I has full column rank, so the final OLS is identified on that event.

The numerator, divided by n , has the exact decomposition

$$\frac{d' M g + d' M u}{n} = \frac{v' u}{n} + \frac{R'_m R_g}{n} + \frac{v' R_g}{n} + \frac{R'_m u}{n} - \frac{v' P u}{n}. \quad (\text{A.31})$$

Each of the last four terms is $o_{\mathbb{P}}(n^{-1/2})$. For the first and fourth remainder terms, respectively,

$$\frac{|R'_m R_g|}{\sqrt{n}} \leq \sqrt{n} \|R_m\|_n \|R_g\|_n = O_{\mathbb{P}}(t_n L_n / \sqrt{n}) = o_{\mathbb{P}}(1), \quad (\text{A.32})$$

$$\frac{|v' P u|}{\sqrt{n}} \leq \sqrt{n} \|P v\|_n \|P u\|_n = O_{\mathbb{P}}(t_n L_n / \sqrt{n}) = o_{\mathbb{P}}(1). \quad (\text{A.33})$$

For the second remainder term, use (A.27) to write

$$\frac{v' R_g}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_i v_i r_{gi} + \frac{v' P_B w_g}{\sqrt{n}}. \quad (\text{A.34})$$

The first term has mean zero because r_{gi} is a function of z_i and $\mathbb{E}[v_i | z_i] = 0$. By independence across observations and Cauchy–Schwarz,

$$\mathbb{E}[v_i^2 r_{gi}^2] \leq (\mathbb{E}[v_i^4])^{1/2} (\mathbb{E}[r_{gi}^4])^{1/2} \leq C s / n \rightarrow 0. \quad (\text{A.35})$$

Hence that centred sum is $o_{\mathbb{P}}(1)$. The second term in (A.34) is at most

$$\sqrt{n} \|P_B v\|_n \|w_g\|_n = O_{\mathbb{P}}(\sqrt{n} a_n^2) = O_{\mathbb{P}}(t_n L_n / \sqrt{n}) = o_{\mathbb{P}}(1). \quad (\text{A.36})$$

The term $R'_m u / \sqrt{n}$ is treated identically, using $\mathbb{E}[u_i | z_i] = 0$, the bound for $P_B u$, and the fourth-moment bound on r_m . These bounds therefore account for the dependence of the selected model on u and v .

Combining (A.30)–(A.36) establishes (3.31). Finally, $\mathbb{E}[v_i u_i] = 0$ follows from $v_i = d_i - m(z_i)$ and $\mathbb{E}[u_i | d_i, z_i] = 0$. For $0 < \delta \leq \min\{1, (q - 4)/2\}$, Hölder's inequality gives

$$\mathbb{E}[|v_i u_i|^{2+\delta}] \leq C. \quad (\text{A.37})$$

Together with $H_n \geq c$, the Lyapunov central limit theorem yields

$$\frac{n^{-1/2} \sum_i v_i u_i}{\sqrt{H_n}} \rightsquigarrow N(0, 1). \quad (\text{A.38})$$

The moment bound supplies the corresponding uniform normal approximation over classes with common constants. It remains to show that the variance can be estimated consistently.

A.2.6 Consistency of the robust variance estimator

Consistency of the corresponding variance estimator for Post-Double Selection is established in Belloni et al. (2014a, Appendix B, proof of Theorem 1, Step 7). For the present assumptions, the fourth-moment approximation bound permits a direct derivation through the empirical fourth moments of the residuals. In particular, the approximation condition and Markov's inequality give

$$\|r_f\|_{n,4} = O_{\mathbb{P}}(\sqrt{s/n}) = o_{\mathbb{P}}(1), \quad f = m, g. \quad (\text{A.39})$$

Apply (A.28) to $w_m, w_g \in \text{span}(X_B)$ and $Pu, Pv \in \text{span}(X_I)$. Their empirical norms are $O_{\mathbb{P}}(a_n)$, so

$$\|w_m\|_{\infty} + \|w_g\|_{\infty} + \|Pu\|_{\infty} + \|Pv\|_{\infty} = O_{\mathbb{P}}(\sqrt{t_n L_n a_n}) = O_{\mathbb{P}}(t_n L_n / \sqrt{n}) = o_{\mathbb{P}}(1). \quad (\text{A.40})$$

Since $R_f = r_f + w_f$, it follows that

$$\|R_m\|_{n,4} + \|R_g\|_{n,4} + \|Pu\|_{n,4} + \|Pv\|_{n,4} = o_{\mathbb{P}}(1). \quad (\text{A.41})$$

The residual identities are

$$\widehat{v} - v = R_m - Pv, \quad (\text{A.42})$$

$$\widehat{u} - u = R_g - Pu - (\widehat{\alpha} - \alpha_0)\widehat{v}. \quad (\text{A.43})$$

Since $\|u\|_{n,4} + \|v\|_{n,4} = O_{\mathbb{P}}(1)$ and $\widehat{\alpha} - \alpha_0 = O_{\mathbb{P}}(n^{-1/2})$, it follows that

$$\|\widehat{v} - v\|_{n,4} + \|\widehat{u} - u\|_{n,4} = o_{\mathbb{P}}(1), \quad \|\widehat{v}\|_{n,4} + \|\widehat{u}\|_{n,4} = O_{\mathbb{P}}(1). \quad (\text{A.44})$$

Hölder's inequality now gives

$$\|\widehat{v}\widehat{u} - vu\|_n \leq \|\widehat{v} - v\|_{n,4} \|u\|_{n,4} + \|\widehat{v}\|_{n,4} \|\widehat{u} - u\|_{n,4} = o_{\mathbb{P}}(1). \quad (\text{A.45})$$

All products here are componentwise. Taking the difference of squared empirical norms proves

$$\mathbb{E}_n[\widehat{v}_i^2 \widehat{u}_i^2] - \mathbb{E}_n[v_i^2 u_i^2] = o_{\mathbb{P}}(1). \quad (\text{A.46})$$

Because $q > 4$, $v_i^2 u_i^2$ has a uniformly bounded moment of some order greater than one. The law of large numbers therefore implies $\mathbb{E}_n[v_i^2 u_i^2] - H_n = o_{\mathbb{P}}(1)$. Equation (A.30) similarly gives $\mathbb{E}_n[\widehat{v}_i^2] - J_n = o_{\mathbb{P}}(1)$. Since J_n is bounded away from zero, $\widehat{V}_n - V_n = o_{\mathbb{P}}(1)$.

Slutsky's theorem and (A.38) prove (3.32) and (3.33). Every remainder bound above is uniform when its underlying assumptions are uniform. The uniform normal approximation and variance consistency then prove (3.34). This completes the proof of Theorem 1. $\square$

The usual HC1, HC2, and HC3 corrections also give asymptotically valid inference when computed from the final selected regression (MacKinnon & White, 1985). HC1's degrees-of-freedom factor tends to one because $|I|/n = o_{\mathbb{P}}(1)$. For HC2 and HC3, let h_{ii} denote the leverage in the regression on (d, X_I) . The projection decomposition, (A.8), and (A.9) give

$$\max_i h_{ii} \leq C \frac{|I|L_n}{n} + \frac{\max_i \widehat{v}_i^2}{\widehat{v}'\widehat{v}} = O_{\mathbb{P}}(t_n L_n/n + n^{-1/2}) = o_{\mathbb{P}}(1), \quad (\text{A.47})$$

where the second term uses $\max_i \widehat{v}_i^2 \leq \{n\mathbb{E}_n[\widehat{v}_i^4]\}^{1/2} = O_{\mathbb{P}}(\sqrt{n})$ and (A.30). The HC2 and HC3 residual multipliers therefore converge uniformly to one, giving the same asymptotic variance. This concerns corrections in the selected regression, rather than a jackknife that repeats variable selection after deleting observations.

A.3 A penalty choice satisfying the conditions

The score condition in Assumption 3 can be obtained under an additional restriction on the tails of the disturbances. Suppose there are known positive constants σ_r , independent of n , such that for $r = m, \ell$ and every $a \in \mathbb{R}$,

$$\mathbb{E}[\exp(ae_{ri}) \mid z_i] \leq \exp(a^2 \sigma_r^2 / 2) \quad \text{almost surely.} \quad (\text{A.48})$$

This conditional sub-Gaussian bound permits heteroscedasticity. Choose

$$\widehat{\psi}_{rj} = \sqrt{\mathbb{E}_n[x_{ij}^2]}, \quad \tau_r = \frac{\sigma_r}{\vartheta} \sqrt{\frac{2 \log(4pn)}{n}}, \quad (\text{A.49})$$

setting $\widehat{\psi}_{rj} = 1$ for any zero column, whose score is zero. Conditional on $(z_1, \dots, z_n)$, independence across observations and (A.48) imply

$$\mathbb{P}\left(\left|\mathbb{E}_n[x_{ij}e_{ri}]\right| > \vartheta \tau_r \widehat{\psi}_{rj} \mid z_1, \dots, z_n\right) \leq 2 \exp\{-n\vartheta^2 \tau_r^2 / (2\sigma_r^2)\} = \frac{1}{2pn}. \quad (\text{A.50})$$

A union bound over the $2p$ pairs (r, j) gives the score event with probability at least $1 - 1/n$. On the design event, (A.8) bounds the column-norm loadings above and away from zero, while $\log(4pn) \asymp L_n$ gives $\tau_r \asymp b_n$. Therefore, Assumption 3 holds. The additional tail condition is used for this construction; the main theorem permits any baseline penalties satisfying Assumption 3, including admissible residual-based loadings (Belloni et al., 2014a).

For an example satisfying all the conditions, take $z_i = x_i \sim N(0, \Sigma_n)$ independently, with covariance eigenvalues bounded above and away from zero. Let $m(x) = x' \beta_{m0}$ and $g(x) = x' \beta_{g0}$ have combined support size at most s , and set $v_i = \sigma_v(x_i) \eta_{vi}$ and $u_i = \sigma_u(x_i) \eta_{ui}$, where the η 's are independent standard normals, independent of X , and the scale functions have common positive lower and finite upper bounds. Set $d_i = m(x_i) + v_i$ and $y_i = \alpha_0 d_i + g(x_i) + u_i$. With bounded $|\alpha_0|$, this gives the sampling, moment, identification, and approximation conditions, and supplies the bounds in (A.48). Gaussian tails give (A.9), while Rudelson and Zhou (2012, Theorem 16 and Corollary 17) give (A.8) for $h_n \rightarrow \infty$ sufficiently slowly that $h_n t_n L_n / n \rightarrow 0$. Such a sequence exists under (A.10). Together with (A.49), this provides a nonempty class satisfying the assumptions while allowing heteroscedasticity and irrelevant controls in the amelioration set.

A.4 The Discrete penalty

For the Discrete penalty, $\omega_j = 0$ on A and one elsewhere. On the design event, X_A has full column rank. For fixed coefficients b on A^c , minimising over the unpenalised coefficients gives $\hat{a}(b) = (X_A' X_A)^{-1} X_A' (q - X_{A^c} b)$. Hence

$$\begin{aligned} & \min_{a,b} \left\{ \frac{1}{2n} \|q - X_A a - X_{A^c} b\|_2^2 + \tau \sum_{j \notin A} \hat{\psi}_j |b_j| \right\} \\ &= \min_b \left\{ \frac{1}{2n} \|M_A q - M_A X_{A^c} b\|_2^2 + \tau \sum_{j \notin A} \hat{\psi}_j |b_j| \right\}. \end{aligned} \quad (\text{A.51})$$

The controls in the amelioration set are therefore accounted for before the remaining selection is made, which explains why the integrated penalty can produce a different selection from an ex-post addition. The identity holds for fixed realised baseline loadings. Since sample residualisation need not preserve independence across observations, the proof uses the original sample.

B Derivations Partial R^2 Parameterization

The following derivation is based off of the supplementary material (Belloni, Chernozhukov & Hansen, 2013). Introduce the partially linear model:

$$y_i = d_i' \alpha_0 + x_i'(c_y \beta_0) + \sigma_y(d_i, x_i) u_i \quad (\text{B.1})$$

$$d_i = x_i'(c_d \eta_0) + \sigma_d(x_i) v_i \quad (\text{B.2})$$

where $X = (x_1, \dots, x_n)'$ is an $(n \times p)$ matrix and $\Sigma = \mathbb{V}(x_i)$. The coefficient vectors η_0 and β_0 each have p entries, and u_i and v_i are standard normal innovations. The scales σ_y and σ_d depend on the error specification of the design. Constants c_d and c_y scale the R^2 in Equations (B.2) and (B.1), respectively.

B.1 First-Stage Parametrization

$$R_d^2 := \frac{\mathbb{V}(\text{Explained Variance in Equation (B.2)})}{\mathbb{V}(\text{Total Variance in Equation (B.2)})} \quad (\text{B.3})$$

$$= \frac{\mathbb{V}(x_i'(c_d \eta_0))}{\mathbb{V}(x_i'(c_d \eta_0) + \sigma_d(x_i) v_i)} \quad (\text{B.4})$$

$$= \frac{c_d^2 \eta_0' \Sigma \eta_0}{c_d^2 \eta_0' \Sigma \eta_0 + \mathbb{V}(\sigma_d(x_i) v_i)} \quad (\text{B.5})$$

which can be rewritten as:

$$c_d = \sqrt{\frac{\mathbb{V}(\sigma_d(x_i) v_i) R_d^2}{(1 - R_d^2) \eta_0' \Sigma \eta_0}} \quad (\text{B.6})$$

B.2 Second-Stage Parametrization

$$R_y^2 := \frac{\mathbb{V}(\text{Explained Variance in Equation (B.1)})}{\mathbb{V}(\text{Total Variance in Equation (B.1)})} \quad (\text{B.7})$$

$$= \frac{\mathbb{V}(x_i'(c_d \eta_0 + c_y \beta_0))}{\mathbb{V}(x_i'(c_d \eta_0 + c_y \beta_0) + \sigma_y(d_i, x_i) u_i)} \quad (\text{B.8})$$

$$= \frac{(c_d \eta_0 + c_y \beta_0)' \mathbb{V}(x_i) (c_d \eta_0 + c_y \beta_0)}{(c_d \eta_0 + c_y \beta_0)' \mathbb{V}(x_i) (c_d \eta_0 + c_y \beta_0) + \mathbb{V}(\sigma_y(d_i, x_i) u_i)} \quad (\text{B.9})$$

where the numerator can be expanded as:

$$(c_d \eta_0 + c_y \beta_0)' \mathbb{V}(x_i) (c_d \eta_0 + c_y \beta_0) = c_d^2 \eta_0' \Sigma \eta_0 + 2c_d c_y \eta_0' \Sigma \beta_0 + c_y^2 \beta_0' \Sigma \beta_0 \quad (\text{B.10})$$

such that

$$R_y^2 = \frac{c_d^2 \eta_0' \Sigma \eta_0 + 2c_d c_y \eta_0' \Sigma \beta_0 + c_y^2 \beta_0' \Sigma \beta_0}{c_d^2 \eta_0' \Sigma \eta_0 + 2c_d c_y \eta_0' \Sigma \beta_0 + c_y^2 \beta_0' \Sigma \beta_0 + \mathbb{V}(\sigma_y(d_i, x_i) u_i)} \quad (\text{B.11})$$

$$0 = (R_y^2 - 1) c_y^2 \beta_0' \Sigma \beta_0 + (R_y^2 - 1) 2c_d c_y \eta_0' \Sigma \beta_0 + (R_y^2 - 1) c_d^2 \eta_0' \Sigma \eta_0 + R_y^2 \mathbb{V}(\sigma_y(d_i, x_i) u_i) \quad (\text{B.12})$$

which can be written as a quadratic model $ax^2 + bx + c = 0$:

$$a = (R_y^2 - 1)\beta_0'\Sigma\beta_0 \quad (\text{B.13})$$

$$b = (R_y^2 - 1)2c_d\eta_0'\Sigma\beta_0 \quad (\text{B.14})$$

$$c = (R_y^2 - 1)c_d^2\eta_0'\Sigma\eta_0 + R_y^2\mathbb{V}(\sigma_y(d_i, x_i)u_i) \quad (\text{B.15})$$

which can be solved using the quadratic formula. Note that the second stage parametrization is directly dependent on the value of c_d , meaning it should be performed in the specified order.

C Code Description

The description of the code is structured according to its affiliation with one of the following categories: General, Replication, Monte Carlo Simulations, Empirical Case, Results, and Computing.

1. General Functions

- (a) `externalFunctions.R`: Includes code sourced from the ‘hdm’ package and supplementary material (Belloni, Chernozhukov & Hansen, 2013; Chernozhukov et al., 2016). Functions include: initial values estimation for the Post-Lasso Iterations, the `LassoShooting` function, and the `Post-Estimator` function.
- (b) `auxFunctions.R`: Contains general functions that are always imported. Functions include: generation of matrices given parameter inputs, partial R^2 parametrization, simulation of $\Lambda(1 - \alpha|X)$, jackknife standard error estimation (MacKinnon & White, 1985), and control selection metrics.
- (c) `auxPenalties.R`: Generates ‘beliefs’ on the amelioration set penalties given a precision level.
- (d) `auxAlgorithms.R`: Contains different Post-Lasso iteration algorithms to estimate σ , conditional on the type of amelioration set penalty and error distribution.

2. Replication Functions

- (a) `mainReplication.R`: A wrapper for all Lasso estimations to run in the replication (Lasso, Post-Lasso, Indirect Post-Lasso, Post-Double Selection, Oracle, Double-Selection Oracle). Collects all the metrics afterwards.
- (b) `auxFunctionsReplication.R`: A subset of functions from `auxFunctions.R` tailored to the replication wrapper.

3. Simulation Functions

- (a) `mainSimulations.R`: A wrapper for all amelioration set penalties simulations to run in the extension.

4. Empirical Case Functions

- (a) `mainExtension.R`: A wrapper for all amelioration set penalties applications to run on different empirical case data.
- (b) `auxFunctionsExtensions.R`: A subset of functions from `auxFunctions.R` tailored to the empirical case.

5. Graphical Functions

- (a) `Results.R`: Generates results based on simulations of the replication and the extension.
- (b) `ResultsCases.R`: Generates results based on different R^2 cases.
- (c) `Results3D.R`: Generates results based on the 3D grid, to be exported with `ExportMATLAB.R`.
- (d) `ExportMATLAB.R`: Exports the 3D grid results to a .csv file, to create graphics in MATLAB.
- (e) Various MATLAB files: Generate different figures using the MATLAB surface function.

6. Computing Functions

- (a) `pcSims.R`: Standard code snippet for distributed computing. For reference, one standard university computer can perform 1000 simulations in approximately 2 hours. Therefore, a (5x5) 3D grid for 3 precision levels, 2 error distributions, and 3 designs takes approximately 900-1000 hours. This can be best distributed among a large number of computers.

D Post-Lasso $\hat{\sigma}$ -Estimation

The iterative Post-Lasso estimation of $\hat{\sigma}$ is adjusted from the methodology of Belloni and Chernozhukov (2011), accommodating the inclusion of Ω . The procedure is applied separately to each selection regression; the index r is suppressed below.

Algorithm 1 Estimation of σ using Post-Lasso iterations

Input: Positive multiplier ψ , tolerance $\nu > 0$, and integer $K \geq 1$

Initialization: Set $k = 0$ and $\hat{\sigma}_0 = \psi \hat{\sigma}_{I_0} > 0$

Use the preliminary OLS residuals as $\hat{e}_0$

while $k < K$ **do**

Set $\lambda_k = c\hat{\sigma}_k\Lambda(1 - \gamma \mid X)$

Set $\hat{\psi}_{j,k} = 1$ under homoscedasticity, or

$\hat{\psi}_{j,k} = \sqrt{\mathbb{E}_n[x_{ij}^2 \hat{e}_{i,k}^2]} / \hat{\sigma}_k$ under heteroscedasticity

Compute the Post-Lasso estimator $\hat{\beta}_k$ using λ_k , $\hat{\Psi}_k$, and Ω_k

Set $\hat{e}_{k+1} = q - X\hat{\beta}_k$ and

$\hat{\sigma}_{k+1} = \sqrt{\hat{Q}(\hat{\beta}_k)} = \sqrt{\hat{e}_{k+1}'\hat{e}_{k+1}/n}$

Set $\Delta_k = |\hat{\sigma}_{k+1} - \hat{\sigma}_k|$ and increment k by one

if $\Delta_{k-1} \leq \nu$ or $\hat{\sigma}_k = 0$ **then**

Break

end if

Update any residual-dependent weights for the next iteration

end while

Output: Report $\hat{\sigma}_k$

The loadings and amelioration weights are held fixed within each minimisation. Under heteroscedasticity, dividing the loadings by $\hat{\sigma}_k$ offsets the same factor in λ_k ; equivalently, the unnormalised loadings in Section 3.5 can be used with $\lambda_k = c\Lambda(1 - \gamma \mid X)$. A zero initial scale or zero loading requires a positive fallback to run the algorithm. The score and loading conditions of the theorem still need to hold for inferential validity.

The multiplier ψ was recommended in the *'hdm'* package to be 0.75. Preliminary tuning has indicated that this value gives similarly strong results for homoscedastic and heteroscedastic data. The constant c is recommended to be 1.1.

E Overview of Ω -penalties

For $j \in \{1, \dots, p\}$ and $r \in \{m, \ell\}$, let $\Omega_{h,r} = (\omega_{h,r1}, \dots, \omega_{h,rp})'$ for each penalty h :

$$\omega_{C,rj} = 1, \tag{E.1}$$

$$\omega_{D,rj} = \begin{cases} 0, & j \in A, \\ 1, & j \notin A, \end{cases} \tag{E.2}$$

$$\omega_{HD,rj} = \begin{cases} 0.5, & j \in A, \\ 1, & j \notin A, \end{cases} \tag{E.3}$$

$$\omega_{A,rj} = \begin{cases} \text{scale}((b_{rh}^{-1})_{h \in A})_j, & j \in A, \\ 1, & j \notin A, \end{cases} \tag{E.4}$$

$$\omega_{S,rj} = \begin{cases} \text{scale}(s_r)_j, & j \in A, \\ 1, & j \notin A. \end{cases} \tag{E.5}$$

Here $b_r = (X_A' X_A)^{-1} X_A' q_r$, $s_r = X_A' \hat{e}_r / n$, and scale is defined in (5.13), with the same fallback conventions. The subscripts are C = Control, D = Discrete, HD = Half-Discrete, A = Adaptive, and S = Score.

F Monte Carlo Simulation Results on Amelioration Set Penalties

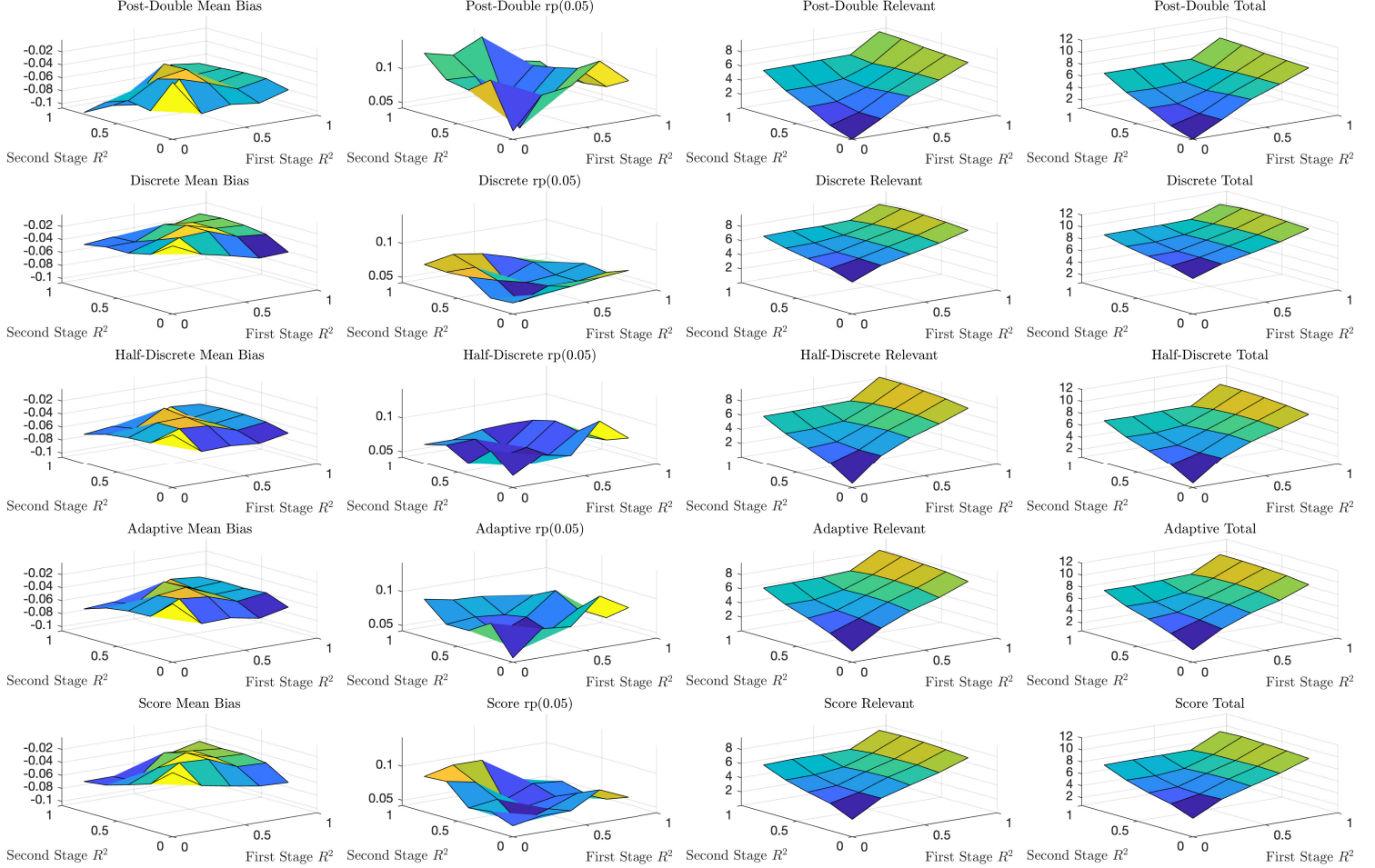


Figure F.1. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 1**, which includes **homoscedastic errors** as detailed in Section 5.1.1.

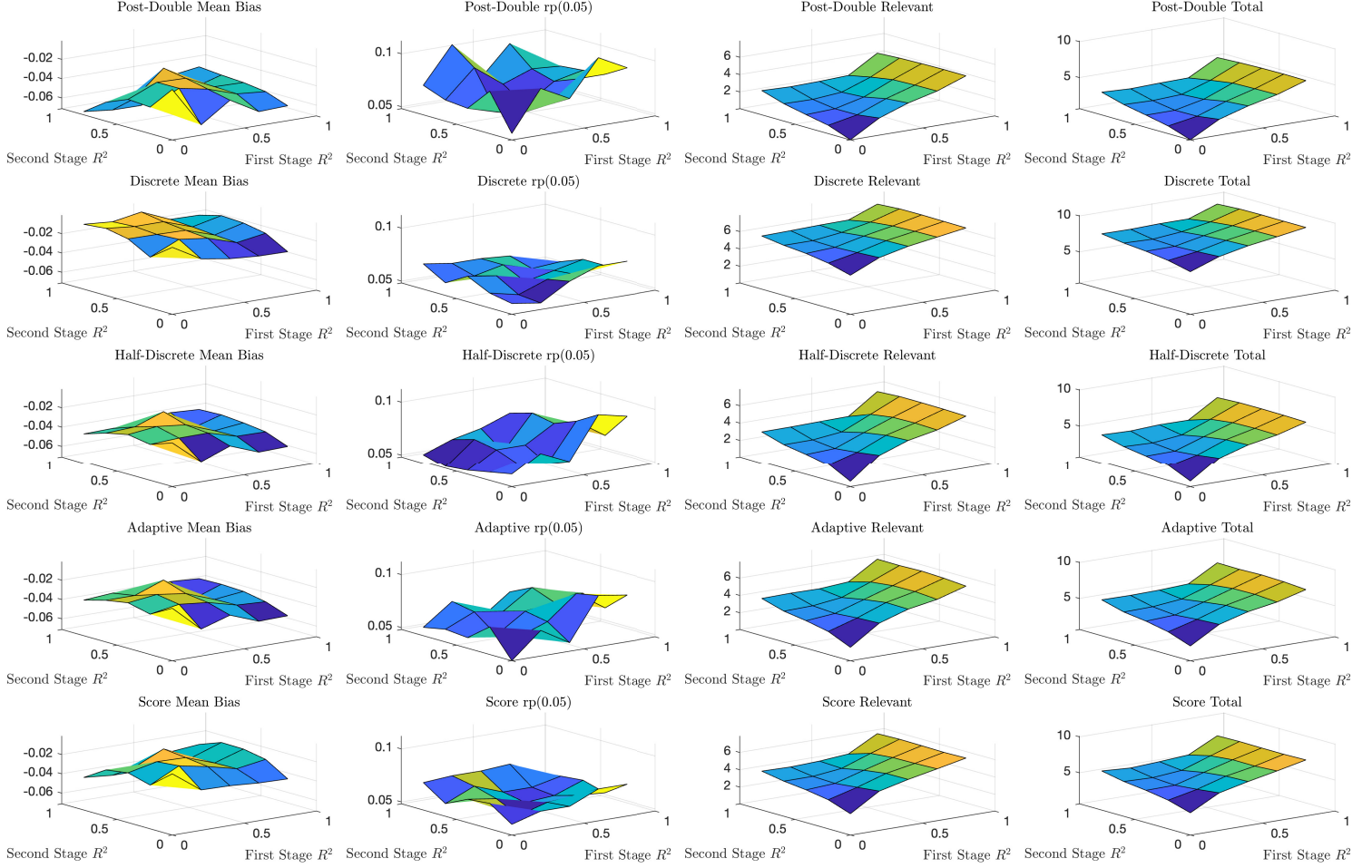


Figure F.2. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 2**, which includes **homoscedastic errors** as detailed in Section 5.1.1.

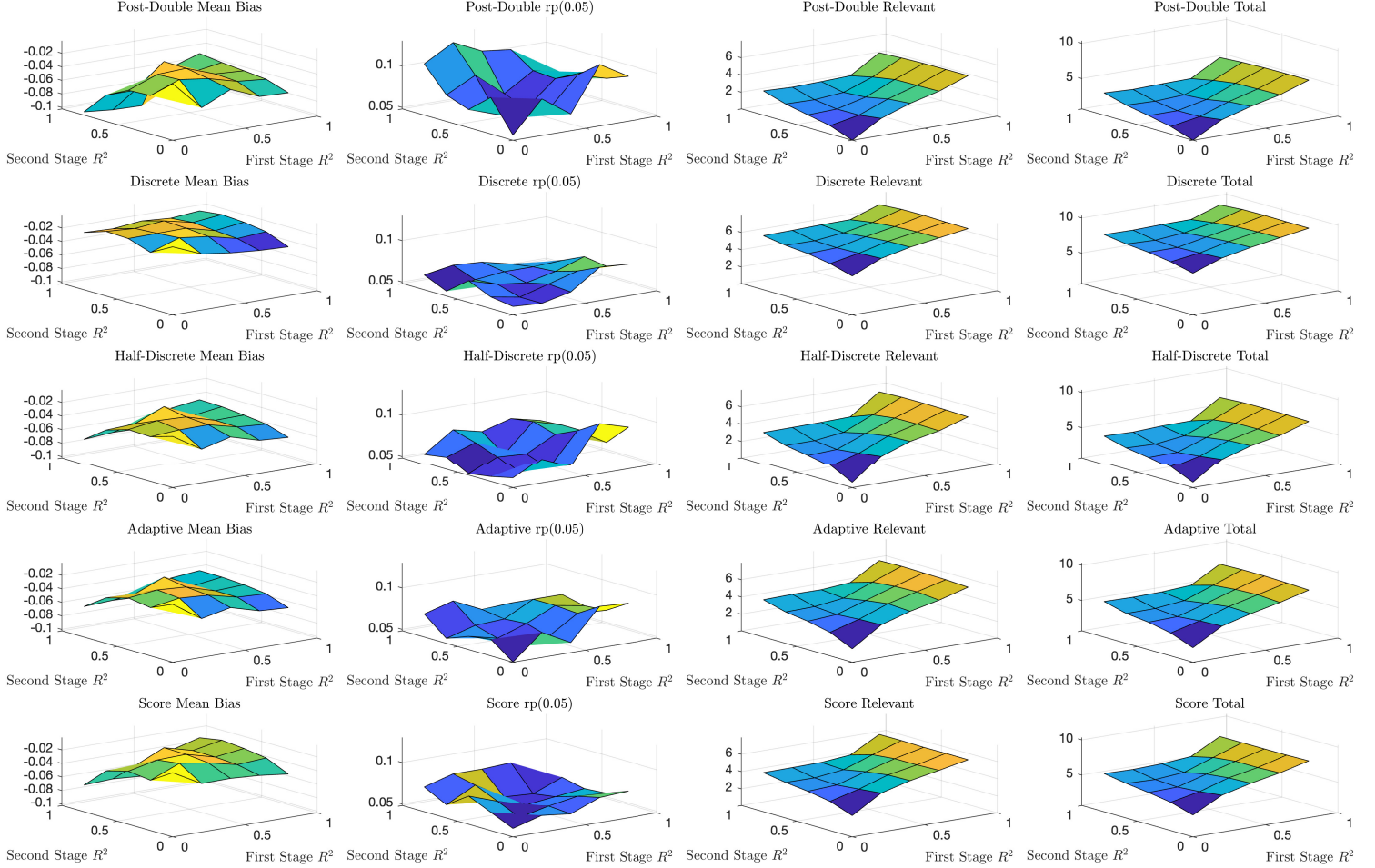


Figure F.3. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 3**, which includes **homoscedastic errors** as detailed in Section 5.1.1.

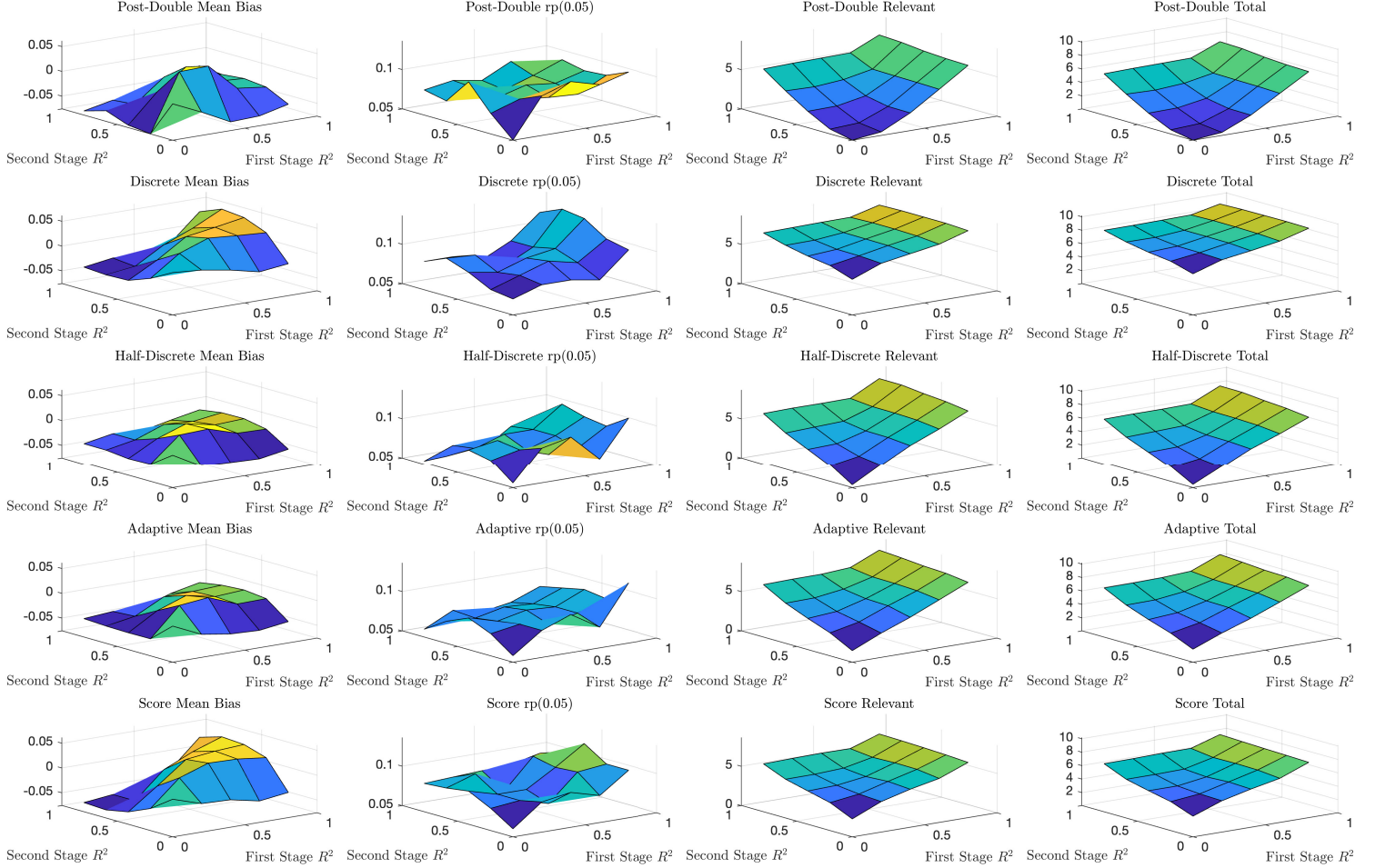


Figure F.4. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 1**, which includes **heteroscedastic errors** as detailed in Section 5.1.1.

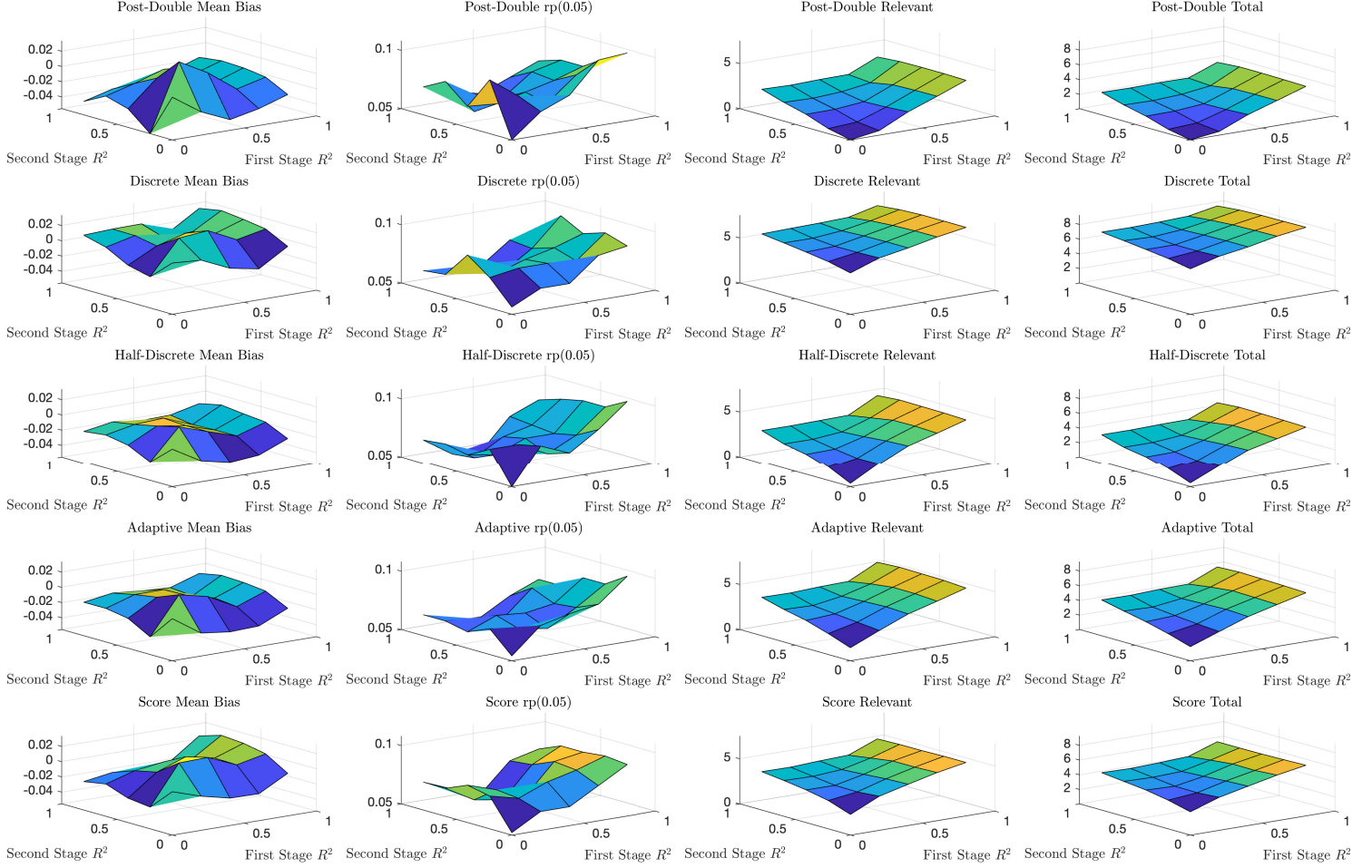


Figure F.5. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 2**, which includes **heteroscedastic errors** as detailed in Section 5.1.1.

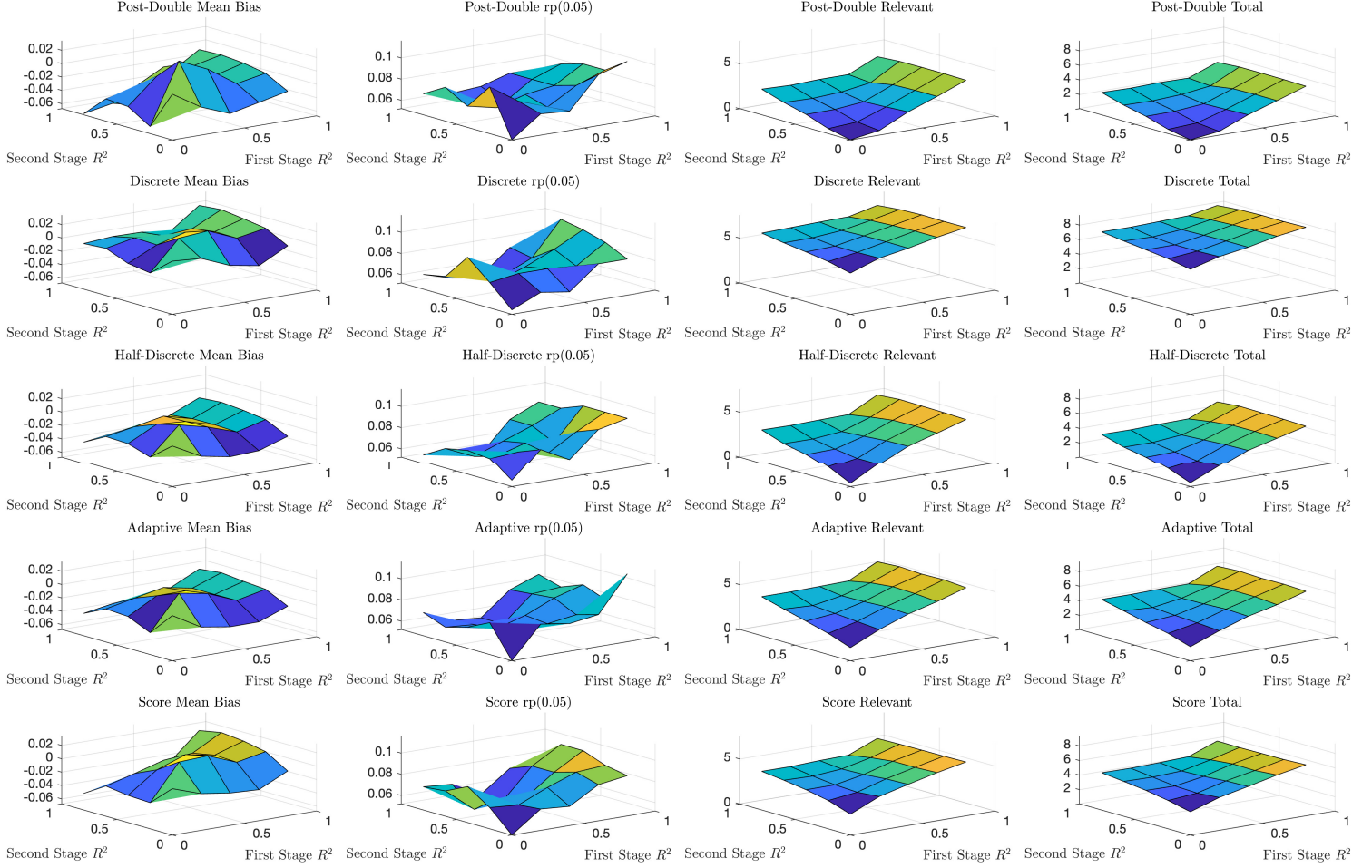


Figure F.6. The figure reports the mean bias, the rejection probabilities of 95% confidence intervals, the number of selected controls in the true DGP, and the total number of selected controls for both the Post-Double Selection (control) and the Post-Double Selection with amelioration set penalties methods. These penalties are described in Section 5.3 and are evaluated across a grid of first-stage and second-stage R^2 . A 'medium' precision level is assumed for the amelioration set penalties. The results are based on 500 simulations of **Design 3**, which includes **heteroscedastic errors** as detailed in Section 5.1.1.

G Empirical Variable Selection

Table G.1. Cardinalities of selected relevant and total controls in the empirical case

	Violence		Property		Murder	
	Relevant	Total	Relevant	Total	Relevant	Total
Control	0	15	0	17	0	8
Control + ex-post Amel	7	20	7	24	7	15
Discrete	6	20	5	16	4	12
Half-Discrete	0	12	1	13	0	8
Adaptive	2	13	2	14	0	8
Score	3	15	2	13	2	10

Note: Relevant and total number selected for the control, the control with the ex-post addition of the amelioration set, and all amelioration set penalties in the empirical case under the assumption of heteroscedastic errors are described in Section 7. Relevant control selection refers to the selection of economically intuitive variables.

Table G.2. Control selection per amelioration set penalty: Violence

Control	Belloni	Control	Control + Amel	Discrete	Half-Discrete†	Adaptive	Score
Lprison	✓	✓	✓	✓	✓	✓	✓
Lur	✓	✓	✓	✓	✓	✓	✓
Dbeer0 × t	✓	✓	✓	✓	✓	✓	✓
incBar	✓	✓	✓	✓	✓	✓	✓
incBar × t	✓	✓	✓	✓	✓	✓	✓
xV0	✓	✓	✓	✓	✓	✓	✓
Dinc0 ² × t	✓						
LprisonBar × t	✓						
Lpolice		✓	✓	✓	✓	✓	✓
Dinc0 × t		✓	✓	✓	✓	✓	✓
Lbeer0 × t ²		✓	✓	✓	✓	✓	✓
Lprison0 ² × t ²		✓	✓	✓	✓	✓	✓
Linc0 × t		✓	✓	✓	✓		✓
Dur		✓	✓	✓			
Dpov		✓	✓	✓			✓
Dinc0		✓	✓	✓			
Linc0		✓	✓		✓	✓	
Dinc			✓	✓		✓	✓
Dbeer			✓	✓		✓	
Dpolice			✓	✓			✓
Dafdc			✓	✓			
Dprison			✓				
Dbeer ² × t ²				✓			✓

Note: Selected controls in the empirical case of the study on the effect of abortion on violent crime, including the control, the control with the ex-post addition of the amelioration set, and all amelioration set penalties under the assumption of heteroscedastic errors, are described in Section 7. The amelioration set is constructed from economically intuitive variables.

Table G.3. Control selection per amelioration set penalty: Property

Control	Belloni	Control	Control + Amel	Discrete	Half- Discrete	Adaptive	Score
Lprison	✓	✓	✓	✓	✓	✓	✓
Dinc0	✓	✓	✓	✓	✓	✓	✓
Linc0	✓	✓	✓	✓	✓	✓	✓
Dbeer0 $\times$ t	✓	✓	✓	✓	✓	✓	✓
incBar	✓	✓	✓	✓	✓	✓	✓
xP0	✓	✓	✓	✓	✓	✓	✓
Linc	✓	✓	✓		✓	✓	
Dinc0 ² $\times$ t	✓	✓	✓				
incBar $\times$ t	✓	✓	✓				
DincBar0 ² $\times$ t	✓	✓	✓				
afdcBar	✓	✓	✓				
afdcBar ²	✓	✓	✓				
Lpolice		✓	✓	✓	✓	✓	✓
Lur		✓	✓	✓	✓	✓	✓
Dinc0 $\times$ t		✓	✓	✓	✓	✓	✓
Lprison0 ² $\times$ t		✓	✓	✓	✓	✓	✓
Lbeer0 ² $\times$ t ²		✓	✓	✓	✓	✓	✓
Dur			✓	✓	✓	✓	✓
Dpolice			✓	✓		✓	
Dpov			✓	✓			✓
Dprison			✓	✓			
Dafdc			✓	✓			
Dbeer			✓				
Dinc			✓				
Lprison0 ² $\times$ t					✓		

Note: Selected controls in the empirical case of the study on the effect of abortion on property crime, including the control, the control with the ex-post addition of the amelioration set, and all amelioration set penalties under the assumption of heteroscedastic errors, are described in Section 7. The amelioration set is constructed from economically intuitive variables.

Table G.4. Control selection per amelioration set penalty: Murder

Control	Belloni	Control	Control + Amel	Discrete	Half- Discrete	Adaptive	Score
Lur	✓	✓	✓	✓	✓	✓	✓
Lprison0 $\times$ t	✓	✓	✓	✓	✓	✓	✓
Dbeer0 $\times$ t ²	✓	✓	✓	✓	✓	✓	✓
incBar $\times$ t	✓	✓	✓	✓	✓	✓	✓
xM0	✓	✓	✓	✓	✓	✓	✓
xM0 $\times$ t	✓	✓	✓	✓	✓	✓	✓
prisonBar $\times$ t	✓						
Dur0 ²	✓						
Lprison	✓						
Linc0 $\times$ t		✓	✓	✓	✓	✓	✓
policeBar $\times$ t		✓	✓	✓	✓	✓	✓
Dpolice			✓	✓			
Dpov			✓	✓			✓
Dbeer			✓	✓			✓
Dafdc			✓	✓			
Dprison			✓				
Dur			✓				
Dinc			✓				

Note: Selected controls in the empirical case of the study on the effect of abortion on murder, including the control, the control with the ex-post addition of the amelioration set, and all amelioration set penalties under the assumption of heteroscedastic errors, are described in Section 7. The amelioration set is constructed from economically intuitive variables.